



REFERENCE MODELS

Network Reference Model - OSI

When Communication is simple

Example 1

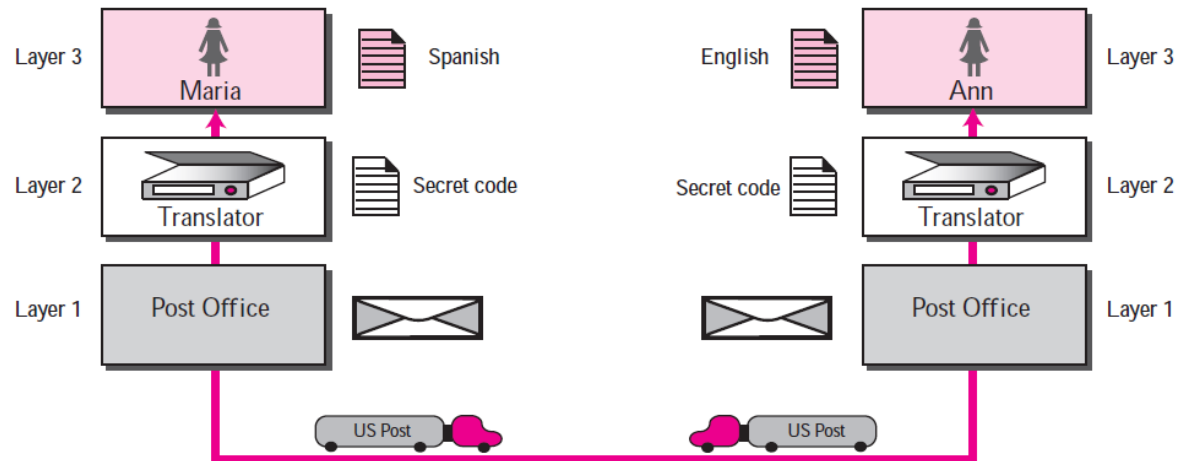
- The communication is so simple that it can occur in only one layer



When Communication is not simple

Example 2

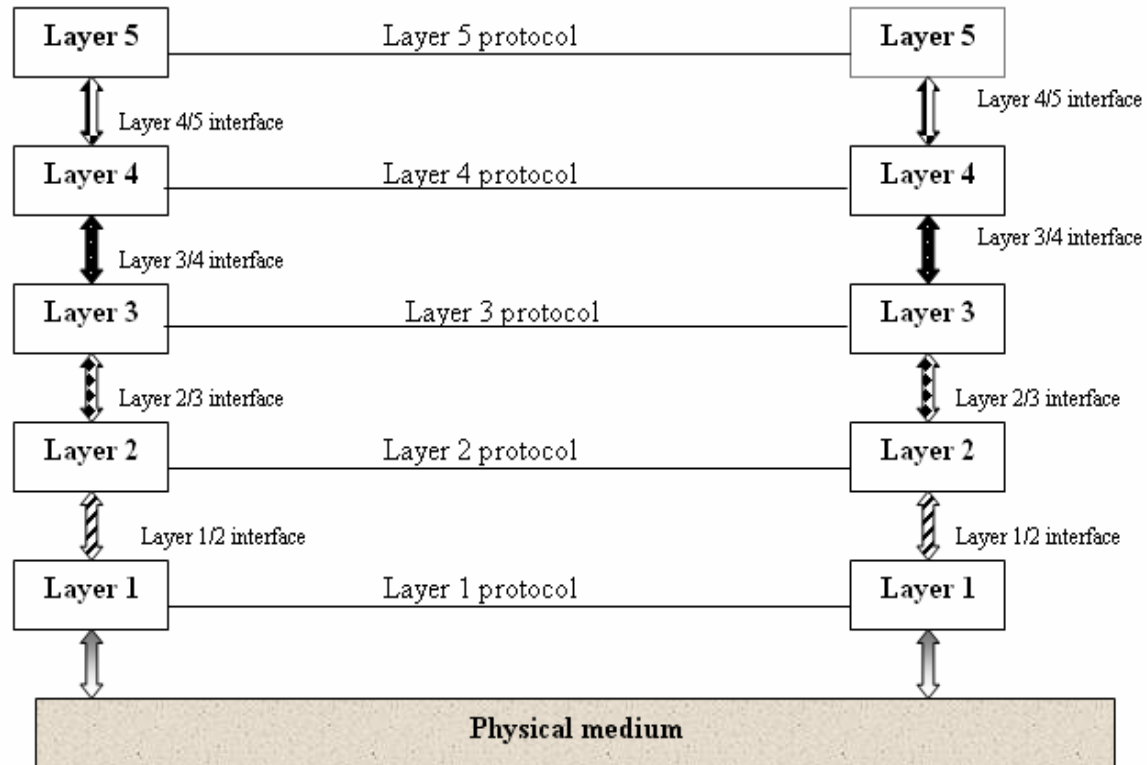
- Ann has to move to another town because of her job
- We need three layers

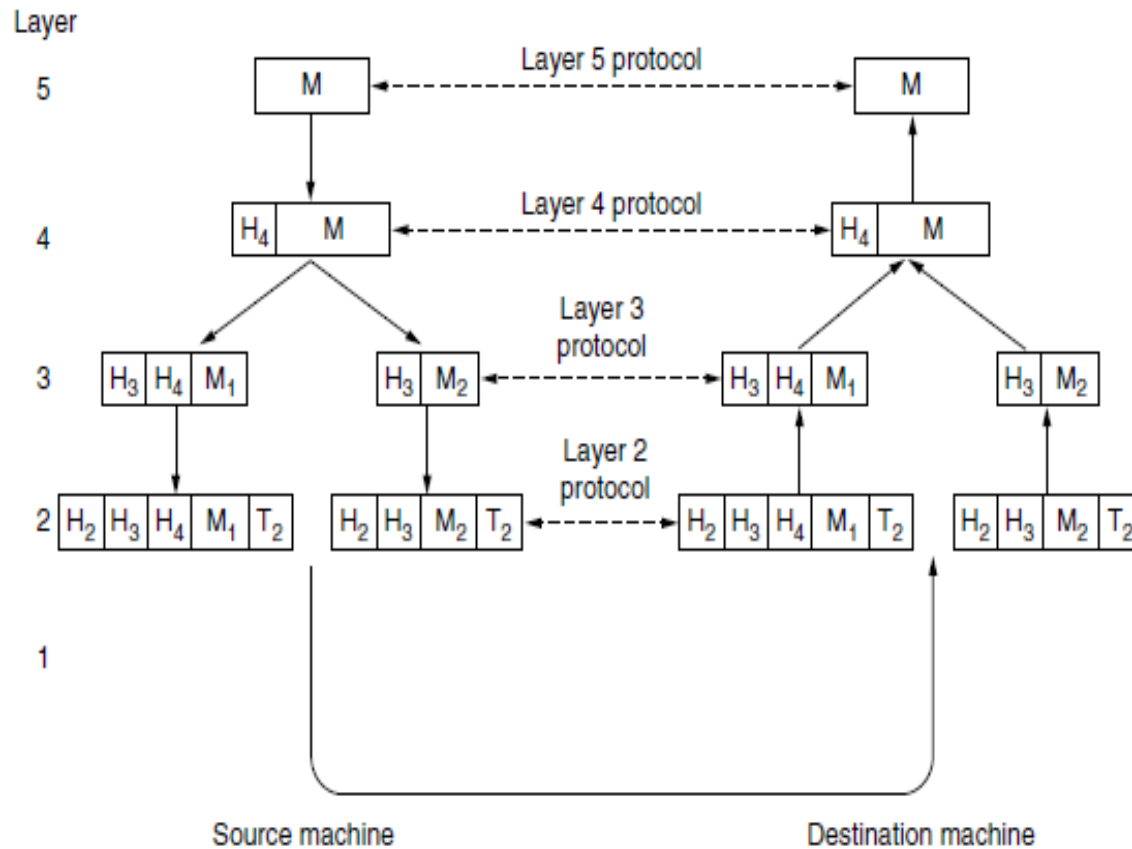


- It is hierarchical (the tasks must be done in the order given in the hierarchy)
- Each layer uses the services of the layer immediately below it
- Each layer gives the services to the layer immediately above it

1. To make the design process easy by **breaking unmanageable tasks into several smaller and manageable tasks** (by **divide-and-conquer** approach).
2. **Modularity and clear interfaces**, so as to provide comparability between the different providers' components.
3. Ensure **independence of layers**, so that implementation of each layer can be changed or modified **without affecting other layers**.
4. **Each layer can be analyzed and tested independently** of all other layers.

- A **service** is a set of actions that a layer offers to another (higher) layer.
- **Protocol** is a set of rules that a layer uses to exchange information with a peer entity. These rules concern both the contents and the order of the messages used.
- Between the layers **service Interfaces** are defined. The messages from one layer to another are sent through those interfaces.







THE OSI REFERENCE MODEL

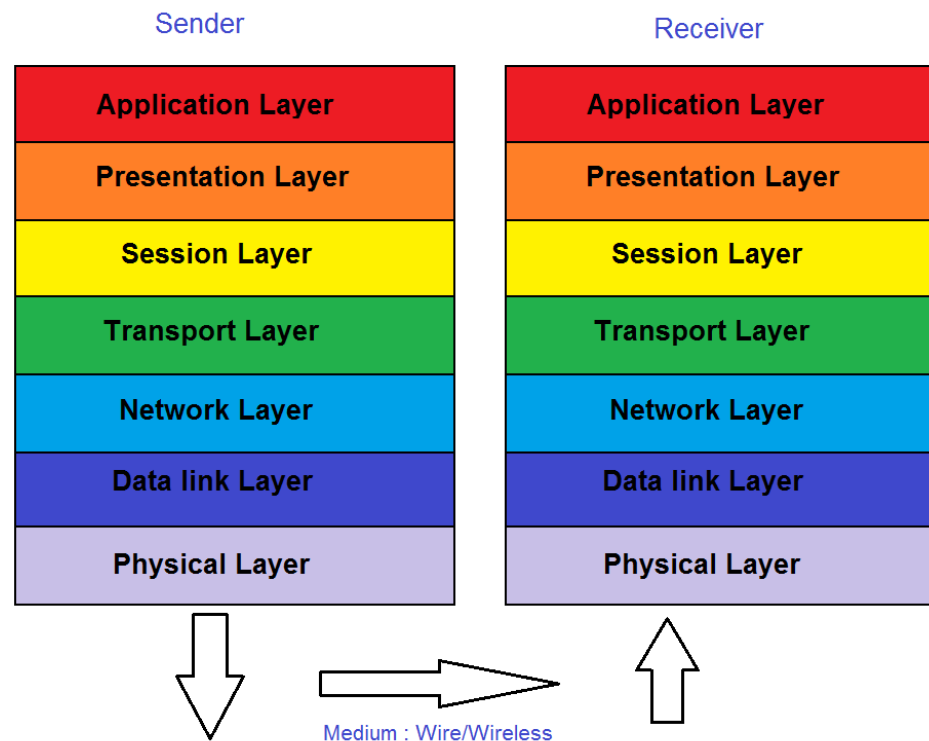


Introduction

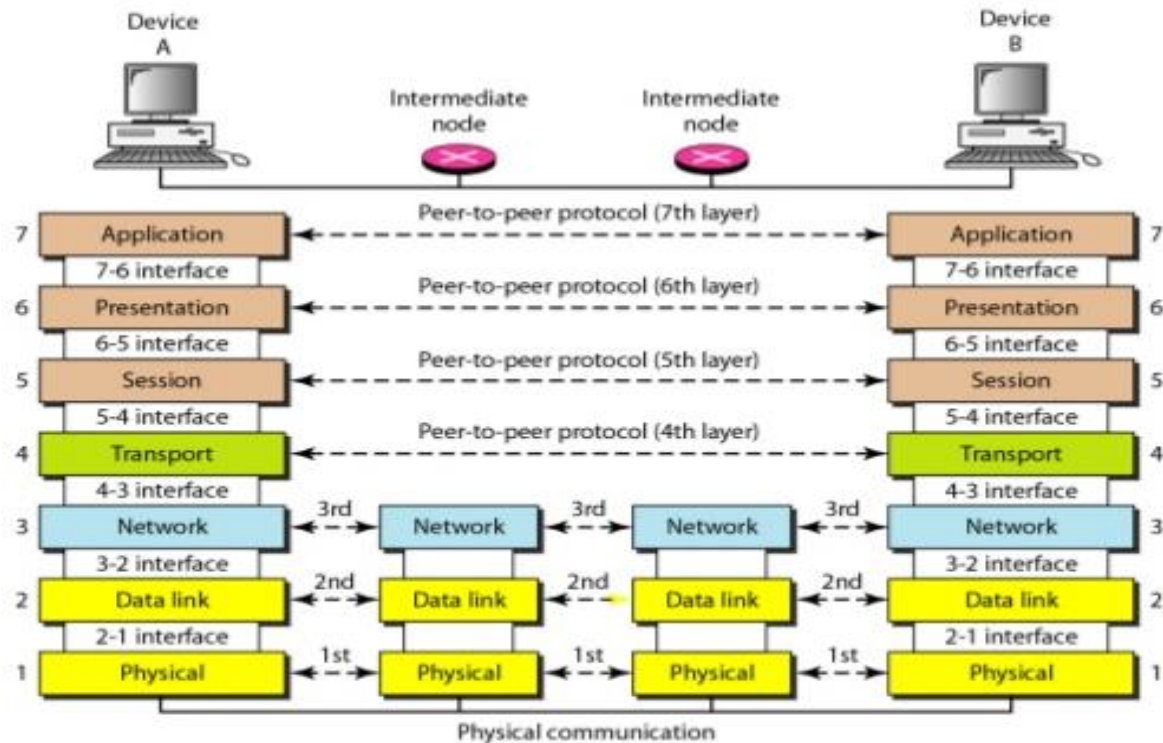
- Open Systems Interconnection Basic Reference Model (OSI Reference Model or **OSI Model**) is an abstract description for **layered communications and computer network protocol design**.
- It was developed as part of the **Open Systems Interconnection (OSI)** initiative.
- In its most basic form, it divides network architecture into seven layers which, from top to bottom, are the **Application, Presentation, Session, Transport, Network, Data-Link, and Physical Layers**.
- It is therefore often referred to as the **OSI Seven Layer Model**.



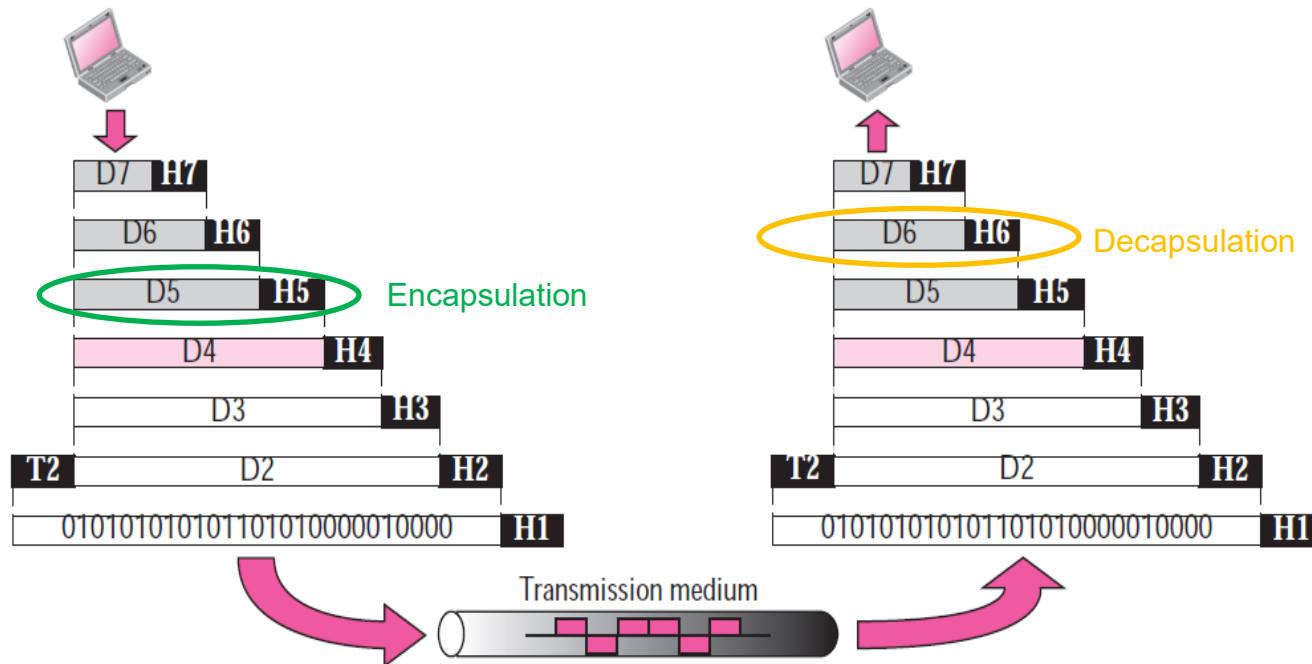
Open System Interconnection (OSI) Model



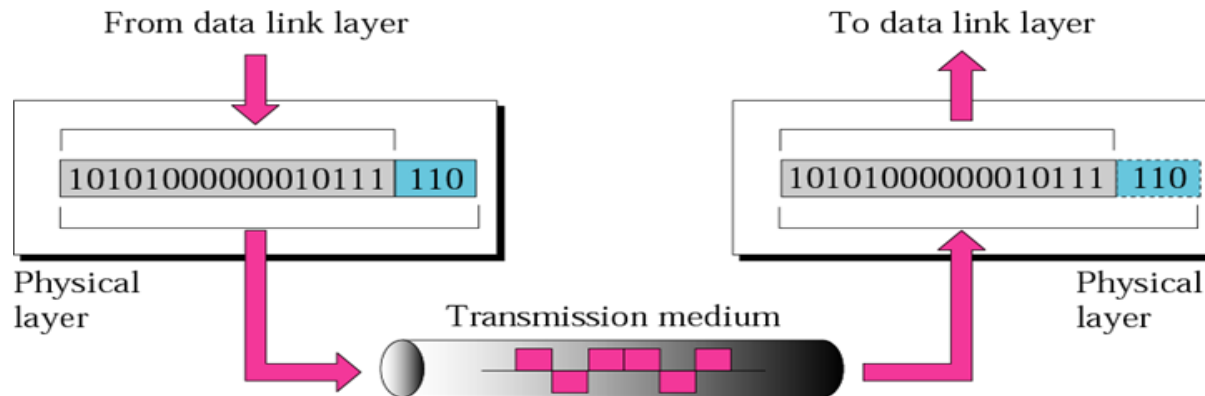
The interaction between the layers in the OSI model



An exchange using the OSI model



1. Physical Layer



- One of the **major function** of the physical layer is to **move data in the form of electromagnetic signals** across a transmission medium.
- Responsible for moving **individual bits** from one (node) to the next.
- It coordinates the functions required to carry a bit stream **over a physical medium which can be wired or wireless**.
- Both data and the signals can be **either analog or digital**.
- It **deals with the mechanical and electrical specifications of the interface and transmission media**.
- It defines the procedures and functions that physical devices and interfaces have to perform for transmission to occur.

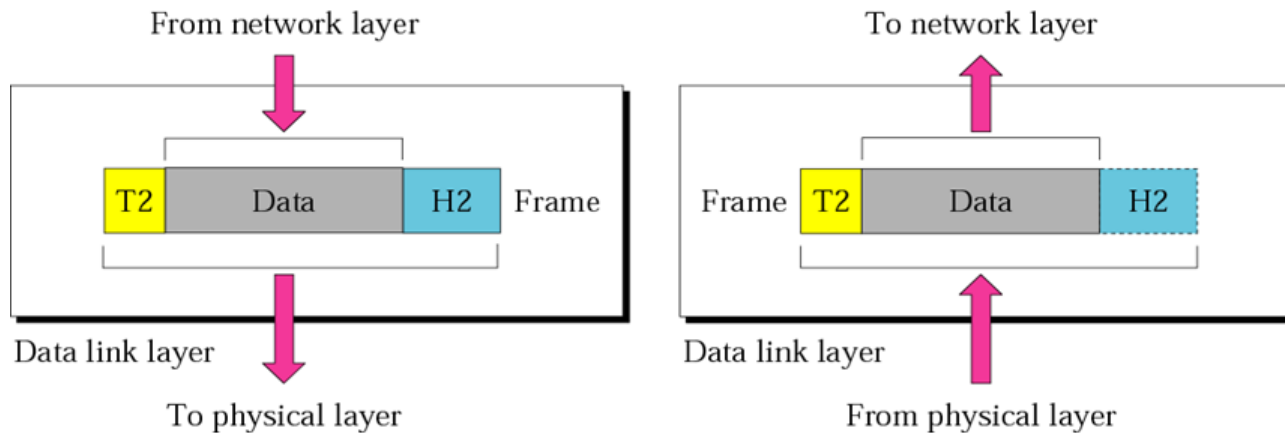
The physical layer is also concerned with the following :



- Physical characteristics of interface and medium (Transmission **medium**)
- Representation of bits (stream of bits (0s or 1s) with no interpretation and **encoded into signals**)
- Data rate
- Synchronization of bits (sender's and receiver's **clock must be synchronized**)
- Line configuration (**Point-to-Point, Point-to-Multipoint**)
- Physical topology
- Transmission mode (**Simplex, half duplex, full duplex**)

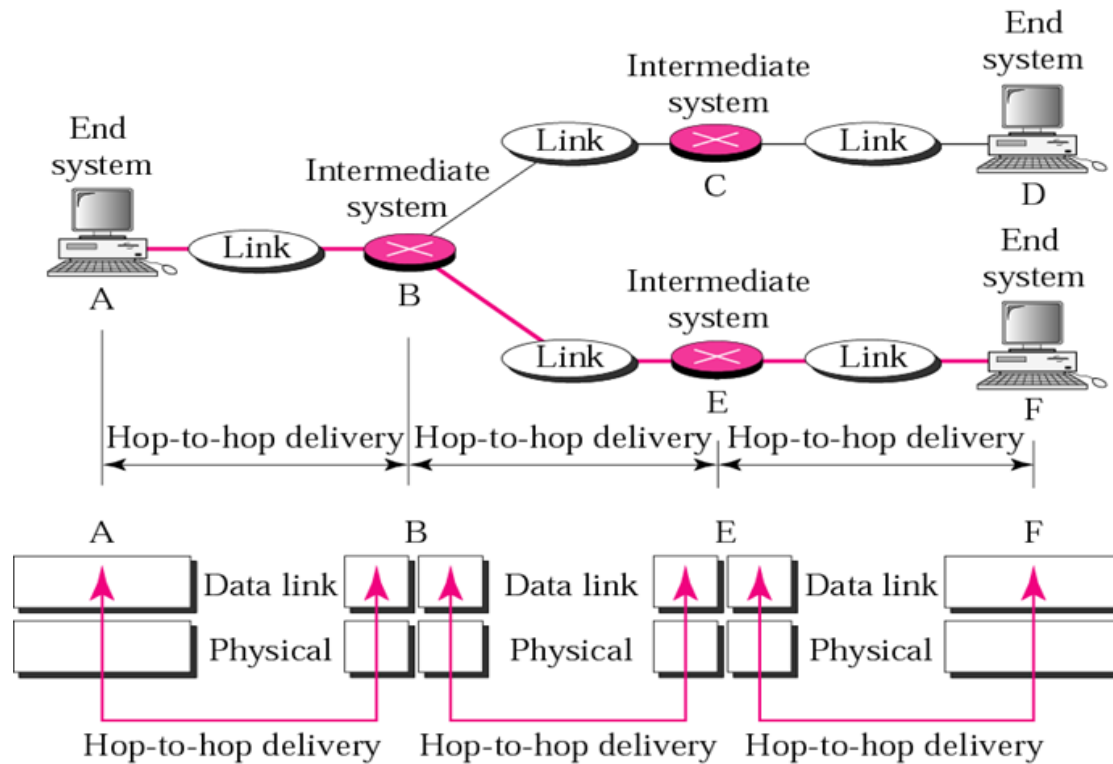
- The physical layer is concerned with transmission of raw bits over a communication channel. It specifies the **mechanical and electrical network interface specifications** and the physical transmission of bit streams over a transmission medium connecting two pieces of communication equipment
- Number of pins and functions of each pin of the network connector (**Mechanical**)
- Signal Level, Data rate (**Electrical**)
- Whether simultaneous transmission in both directions
- Establishing and breaking of connection
- Deals with physical transmission.

2. Data Link Layer



- The data link layer is responsible for *moving frames* from one hop (node) to the next
- The data link layer transforms the *physical layer, a raw transmission facility, to a reliable link*
 - It makes the physical layer appear error-free to the upper layer (network layer)

Hop-to-Hop delivery



Other responsibilities of the data link layer include the following



- Framing (stream of bits into manageable data units)
- Physical addressing (MAC Address)
- Flow Control (mechanism for overwhelming the receiver)
- Error Control (trailer, retransmission)
- Access Control (defining master device in the same link)

1. Group the **physical layer bit stream** into units called frames.
(Note that frames are nothing more than ``packets" or ``messages". By convention, *'frames' term is used when discussing DLL packets.*)
2. Sender **calculates the checksum and sends checksum together with the data.**
(The checksum allows the receiver to determine when a frame has been damaged in transit or received correctly. This is called **error detection and correction.**)

3. Error control protocol returns a positive or negative acknowledgment to the sender.

(A positive acknowledgment indicates the frame was received without errors, while a negative acknowledgment indicates the opposite.)

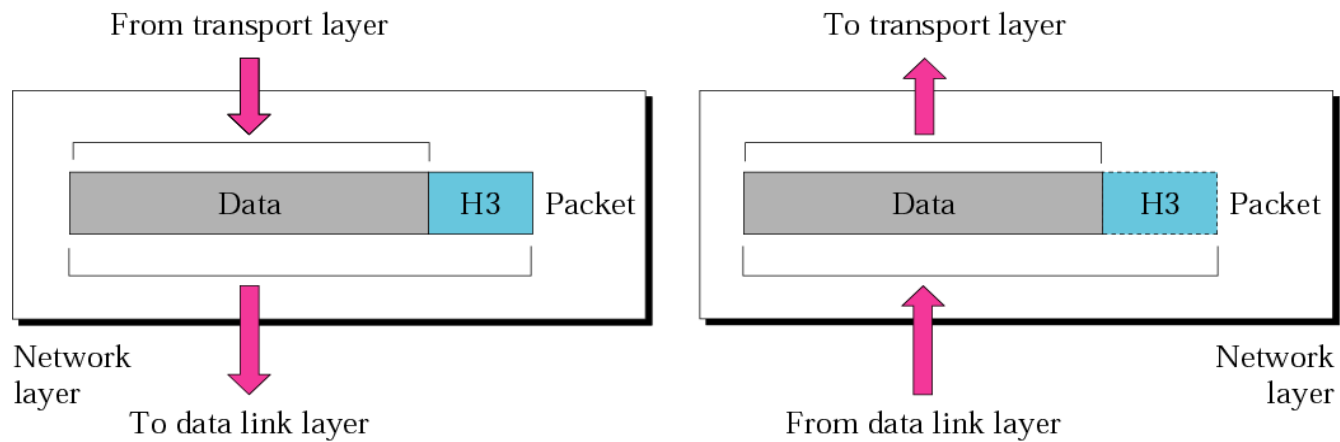
4. Flow control prevents a fast sender from overwhelming a slower receiver.

(For example, a supercomputer can easily generate data faster than a PC can consume it.)

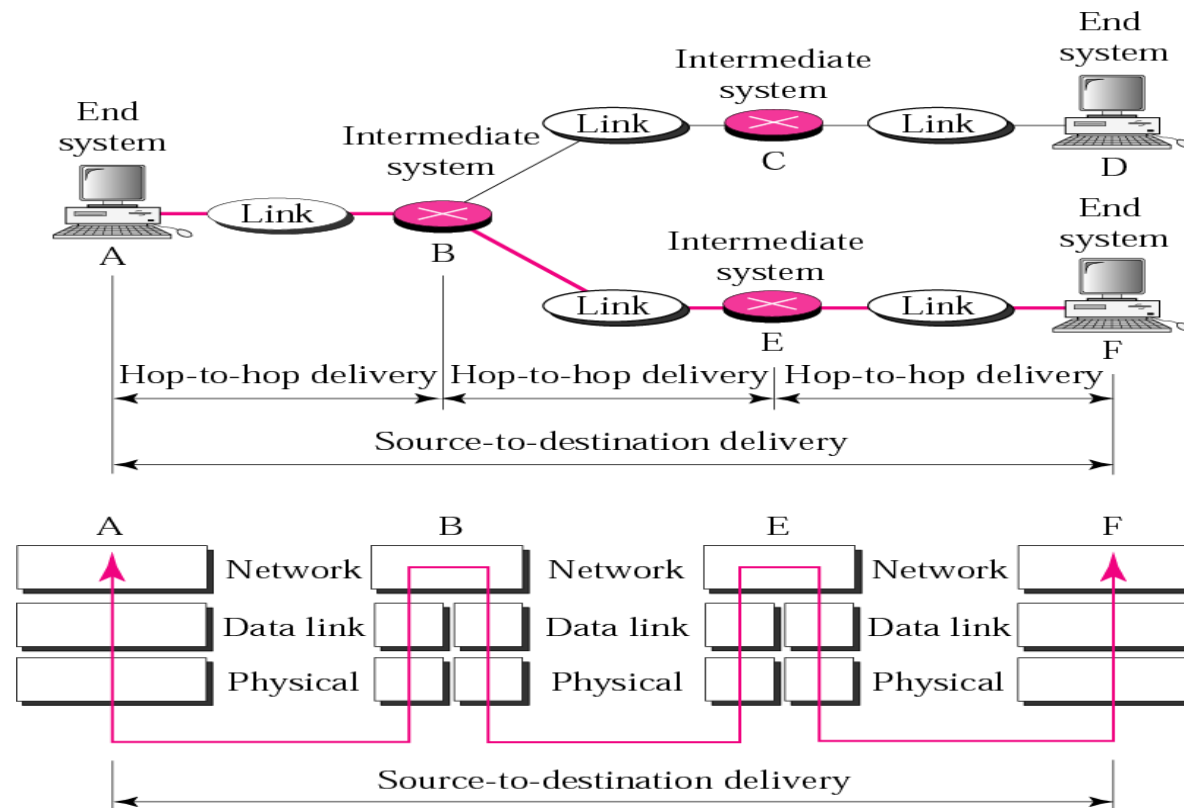
3. Network Layer

- The network layer is responsible for the *source-to-destination* delivery of a *packet*, *possibly across multiple networks* (links)
- Whereas the *data link layer* oversees the delivery of the packet *between two systems on the same network* (link)
- The network layer ensures that each packet gets from its point of origin to its final destination

- If two systems are connected to the **same link**, there is **usually no need for a network layer**
- However, if the two systems are attached to different networks (links) with connecting devices between the networks (links), there is often a need for the network layer to **accomplish source-to-destination delivery**



source-to-destination delivery



Other responsibilities of the network layer include the following

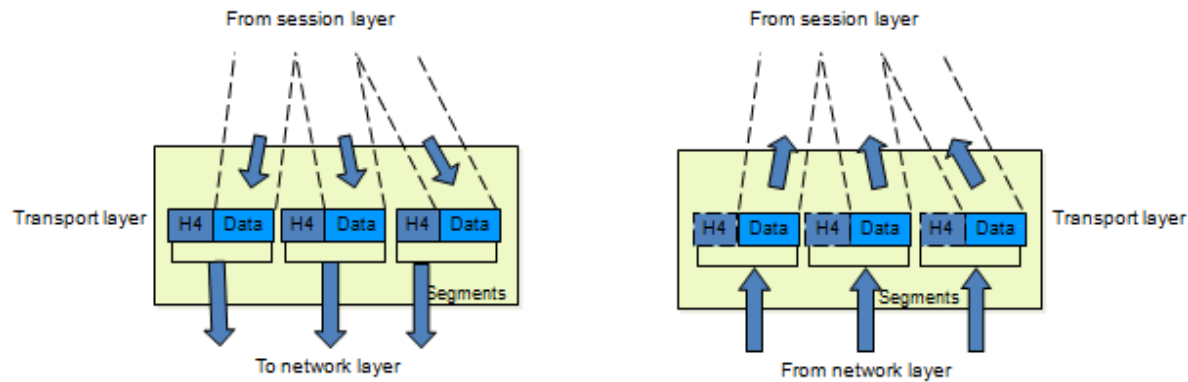


- ☐ Logical addressing (**IP Address**)
- ☐ Routing (Source to destination transmission between networks)

1. Interface between the host and the network (the network layer is typically the boundary between the host and subnet)
2. Routing
3. Congestion and deadlock
4. Internetworking (A path may traverse different network technologies (e.g., Ethernet, point-to-point links, etc.)

4. Transport Layer

- The transport layer is responsible for the **delivery of a message** from one process to **another**
- A **process** is an **application program** running on the host

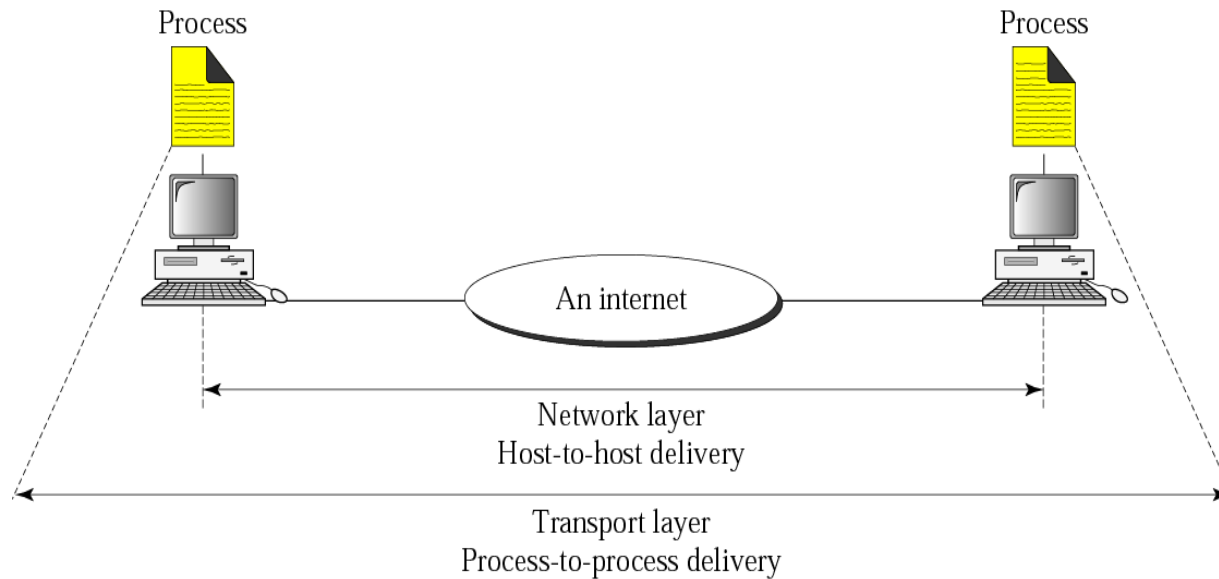


Other responsibilities of the transport layer include the following



- ☐ Service-point addressing (**Port address**)
- ☐ Segmentation and reassembly (**Sequence number**)
- ☐ Connection control (**Connectionless or connection oriented**)
- ☐ Flow control (**end to end**)
- ☐ Error Control (**Process to Process**)

Reliable process-to-process delivery of a message

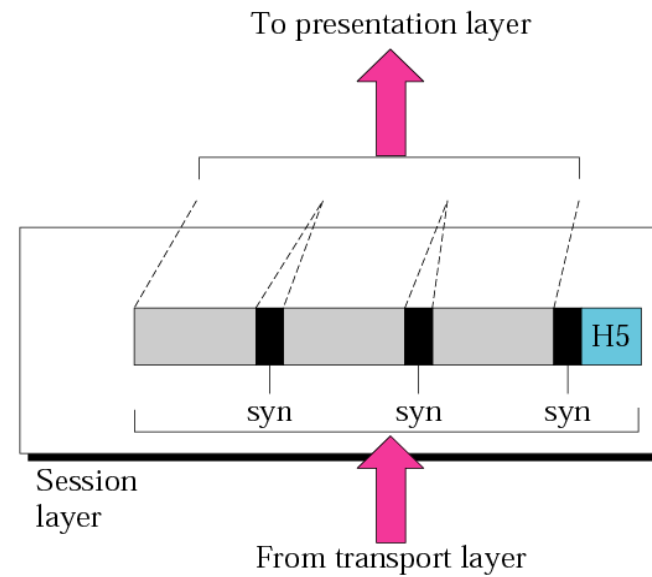
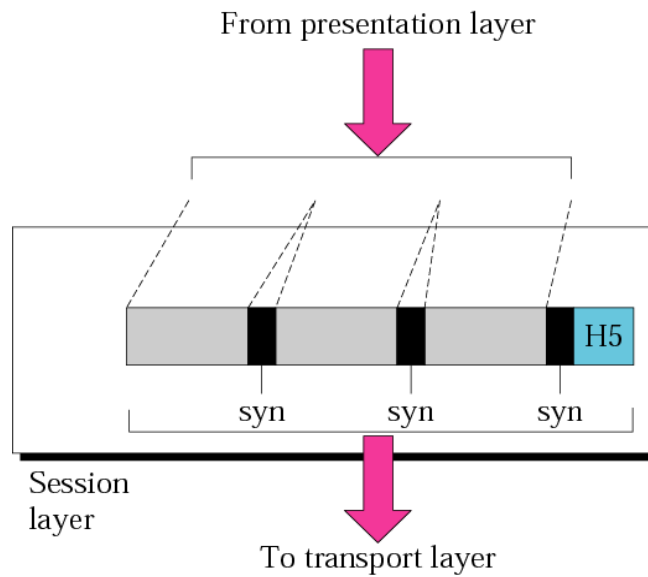


- Network layer oversees *source-to-destination delivery* of *individual packets*
 - It does not recognize any relationship between those packets
- The transport layer ensures that the *whole message arrives intact and in order*
 - Overseeing both error control and flow control at the source-to-destination level

- The transport level provides end-to-end communication between processes executing on different machines.
- Negotiation of Quality and Type of Services.
- Deal with congestion control
- Connection establishment and termination.
- Multiplexing – Demultiplexing.
- Error control

5. Session Layer

- Session layer **establishes, maintains, and synchronizes the interaction** between communicating systems
- Responsibilities
 - Dialog Control (**Half Duplex/Full duplex**)
 - Token Management (to **avoid same operation at the same time**)
 - Synchronization (**Synchronization points**, process inline within same page)



- **Dialogue Control:** Session can allow traffic to go in both direction at the same time, or in only one direction at one time.
- **Token management:** For some protocols, it is required that both sides don't attempt same operation at the same time. To manage these activities, the session layer provides tokens that can be exchanged. Only one side that is holding token can perform the critical operation.
- **Synchronization/Check Pointing:** Session layer provides a way to insert checkpoints into data streams, so that after a crash, only the data transferred after the last checkpoint have to be repeated.

6. Presentation Layer

- The presentation layer is concerned with the syntax and semantics of the information exchanged between two systems.
- The presentation layer is responsible for translation, compression and encryption.
- The responsibilities of this layer is as follows:
 - Translation (interoperability between different encoding system)
 - Encryption (Privacy schemes)
 - Compression (data compression)

- **Data formatting and conversion:**

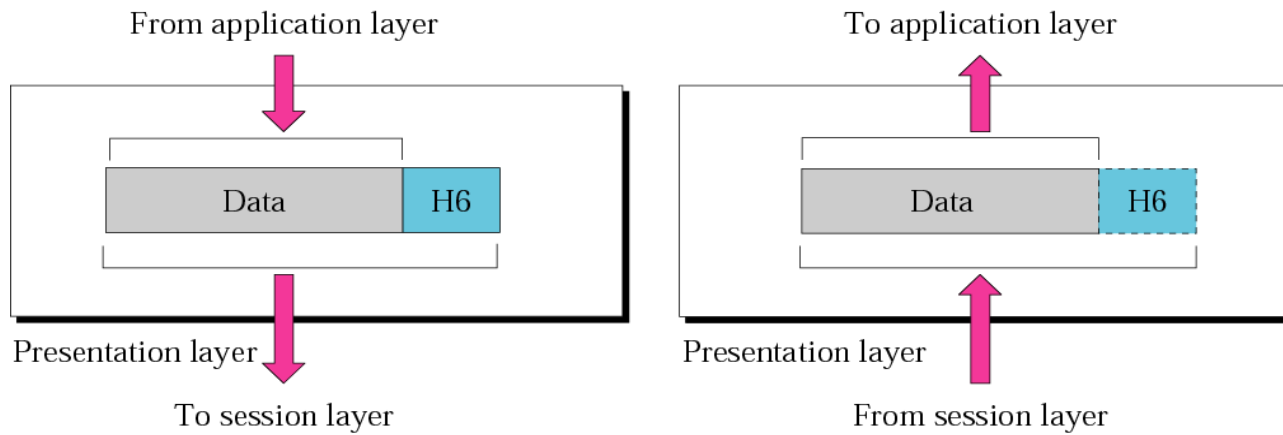
Ensuring that data sent from a *Windows PC* can be correctly interpreted by a *Mac or Unix system*.

- **Encryption and decryption:**

When you access a *secure website* (HTTPS), the layer encrypts the data before sending it and decrypts it upon receipt.

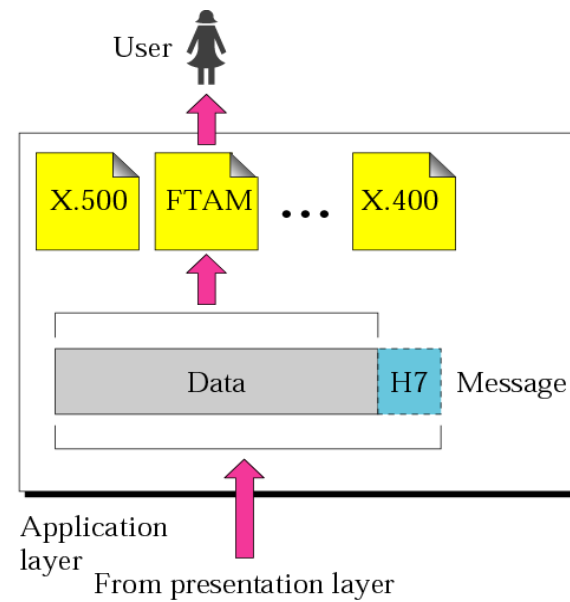
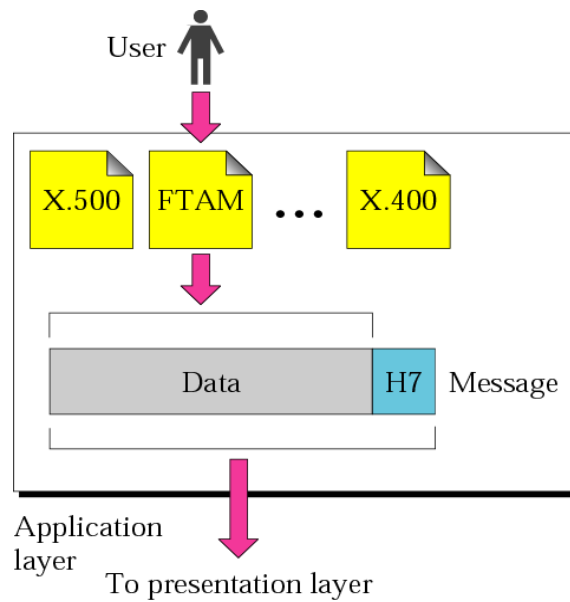
- **Data compression:**

When sending a *large file attachment in an email*.



7. Application Layer

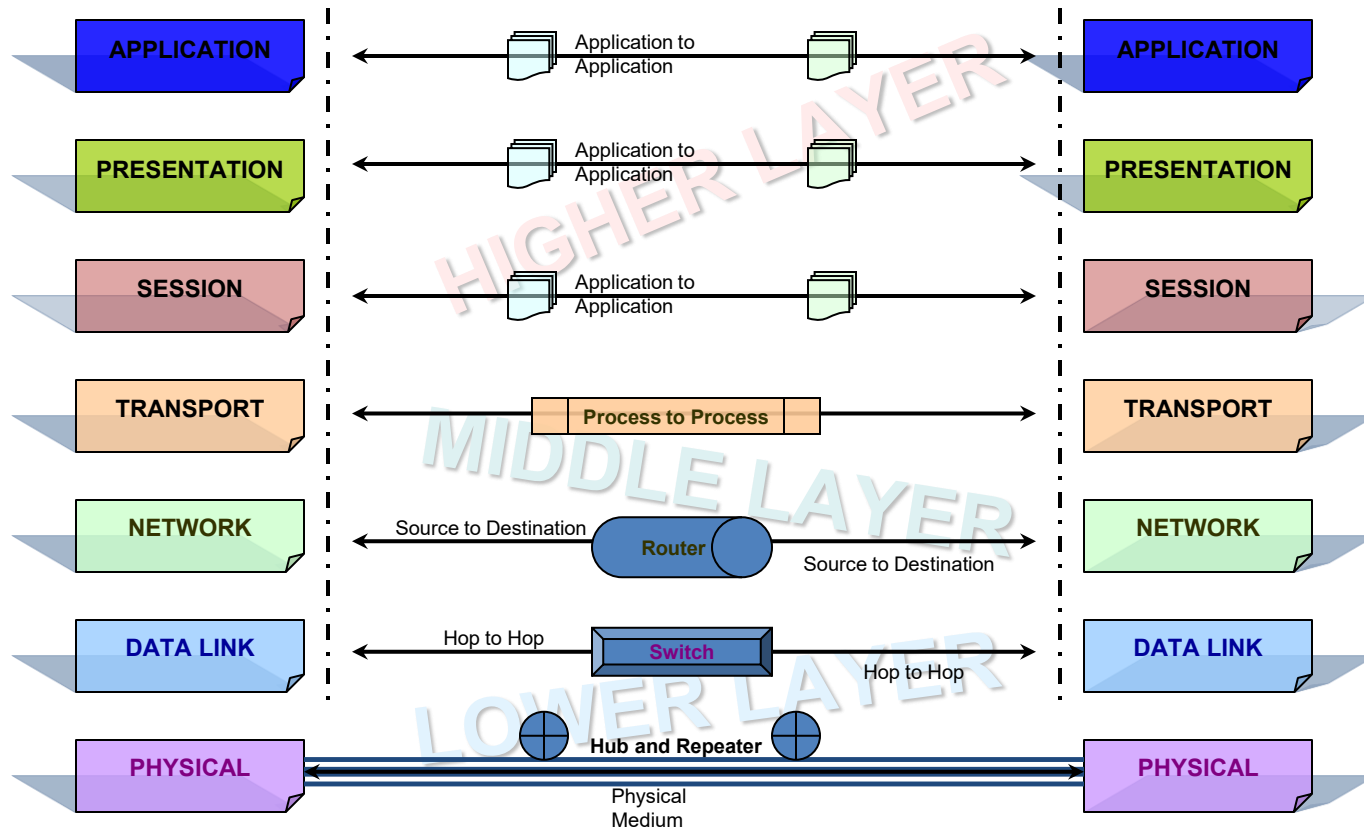
- The application layer enables the user, whether human or software, to access the network
- It provides user interfaces and support for services (email, remote file access etc.)
- Services provided :
 - Network virtual terminal (Software)
 - File transfer, access, and management (FTAM)
 - E-mail services
 - Directory services (access to distributed database sources for global information about various objects and services)



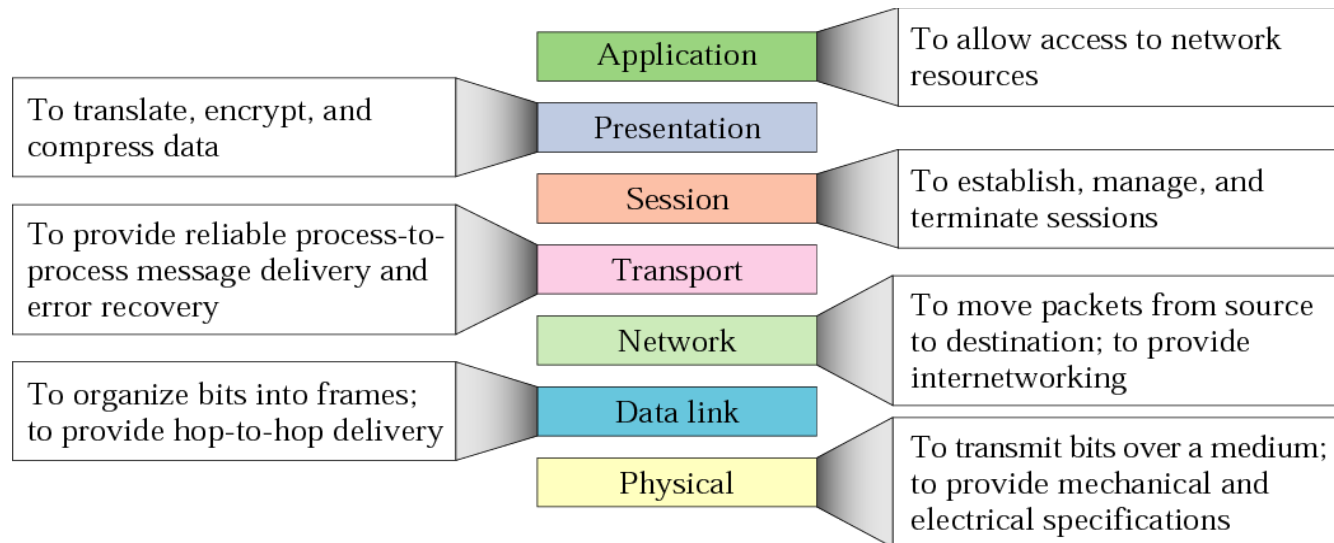


- **File transfer (FTP):** Connect to a remote machine and send or fetch an arbitrary file. FTP deals with authentication, listing a directory contents, ASCII or binary files, etc.
- **Remote login (telnet):** A remote terminal protocol that allows a user at one site to establish a TCP connection to another site, and then pass keystrokes from the local host to the remote host.
- **Mail (SMTP):** Allow a mail delivery agent on a local machine to connect to a mail delivery agent on a remote machine and deliver mail.
- **News (NNTP):** Allows communication between a news server and a news client.
- **Web (HTTP):** Base protocol for communication on the World Wide Web.
- **Domain Name System (DNS):** Translates human-readable domain names into IP addresses.

OSI Model's 7 Layers

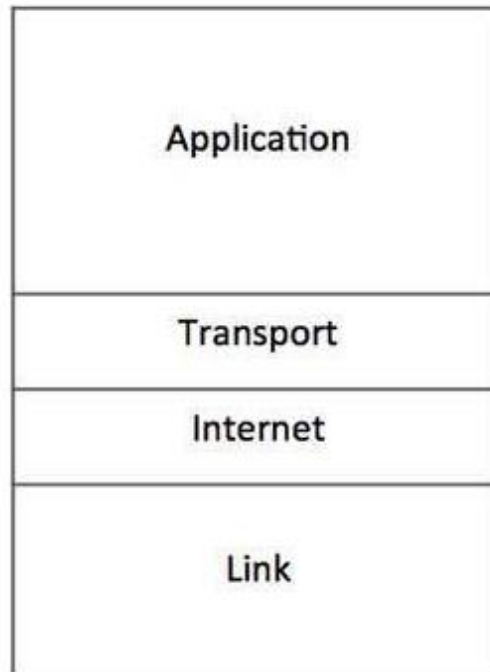


Summary of OSI Layers



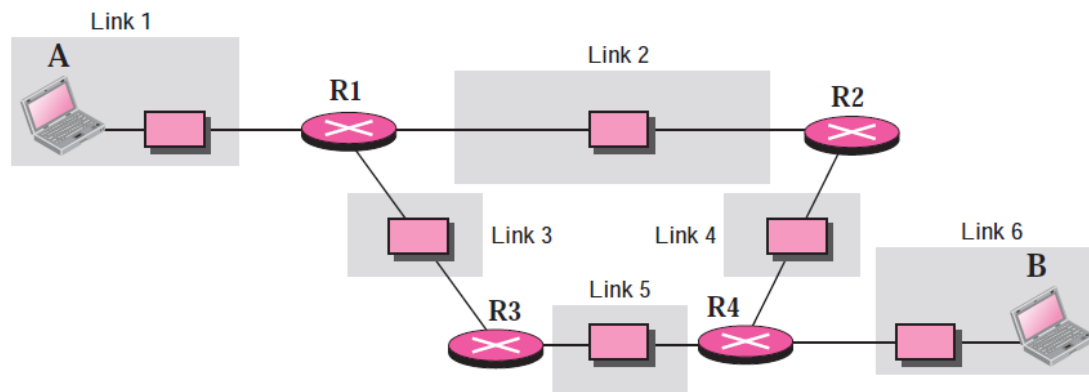


TCP/IP REFERENCE MODEL



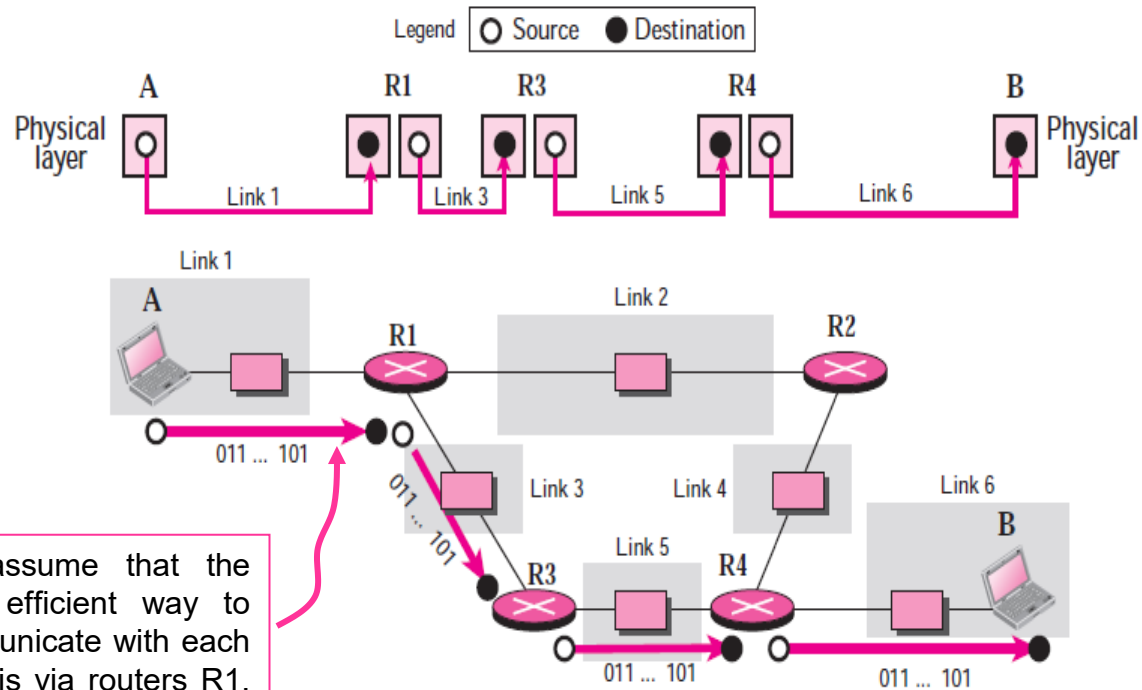
TCP/IP Reference Model

- A link is a network that allows a set of computers to communicate with each other
- A link can be a LAN (local area network) serving a small area or a WAN (wide area network) serving a larger area
- We assume that different links are connected together by devices called *routers* or *switches* that route the data to reach their final destinations



1. Physical Layer

- TCP/IP does **not define any specific protocol for the physical layer**
- It supports all of the standard and proprietary protocols
- The communication is between two hops or nodes, either a computer or router
- The unit of communication is a single ***bit***
- The responsibility of the physical layer, in addition to delivery of bits, matches with what mentioned for the physical layer of the OSI model

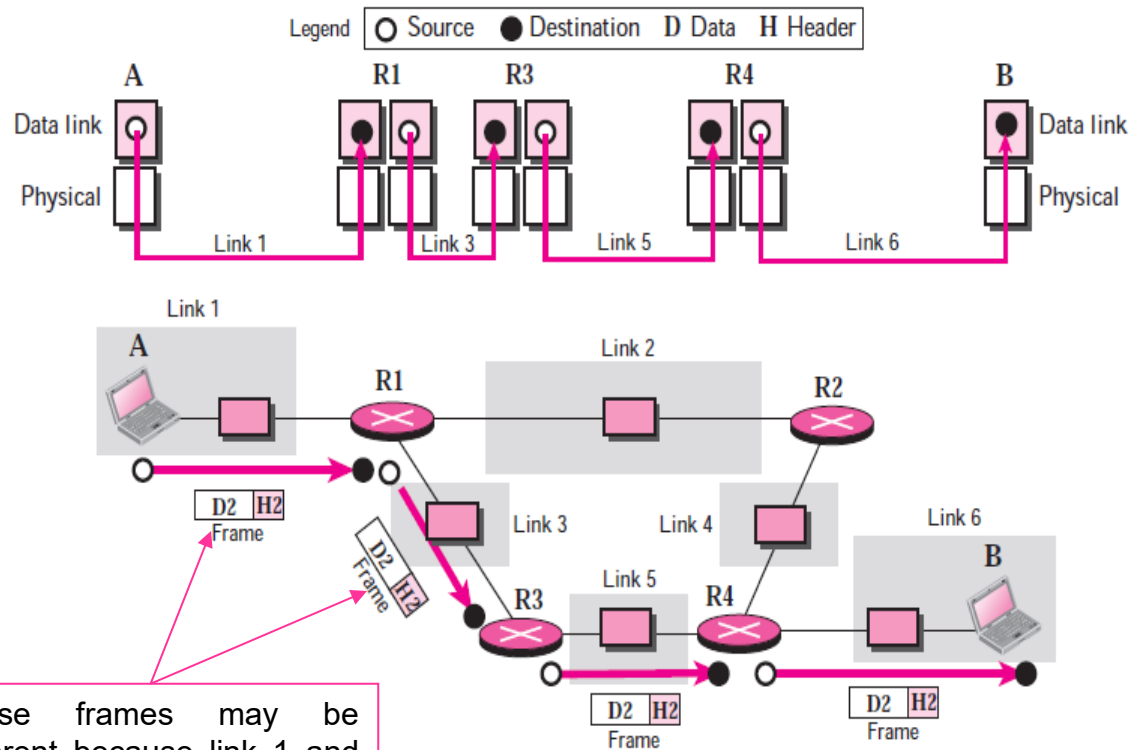


We assume that the most efficient way to communicate with each other is via routers R1, R3, and R4.

if a node is connected to n links, it needs n physical-layer protocols, one for each link because links may use different physical-layer protocols

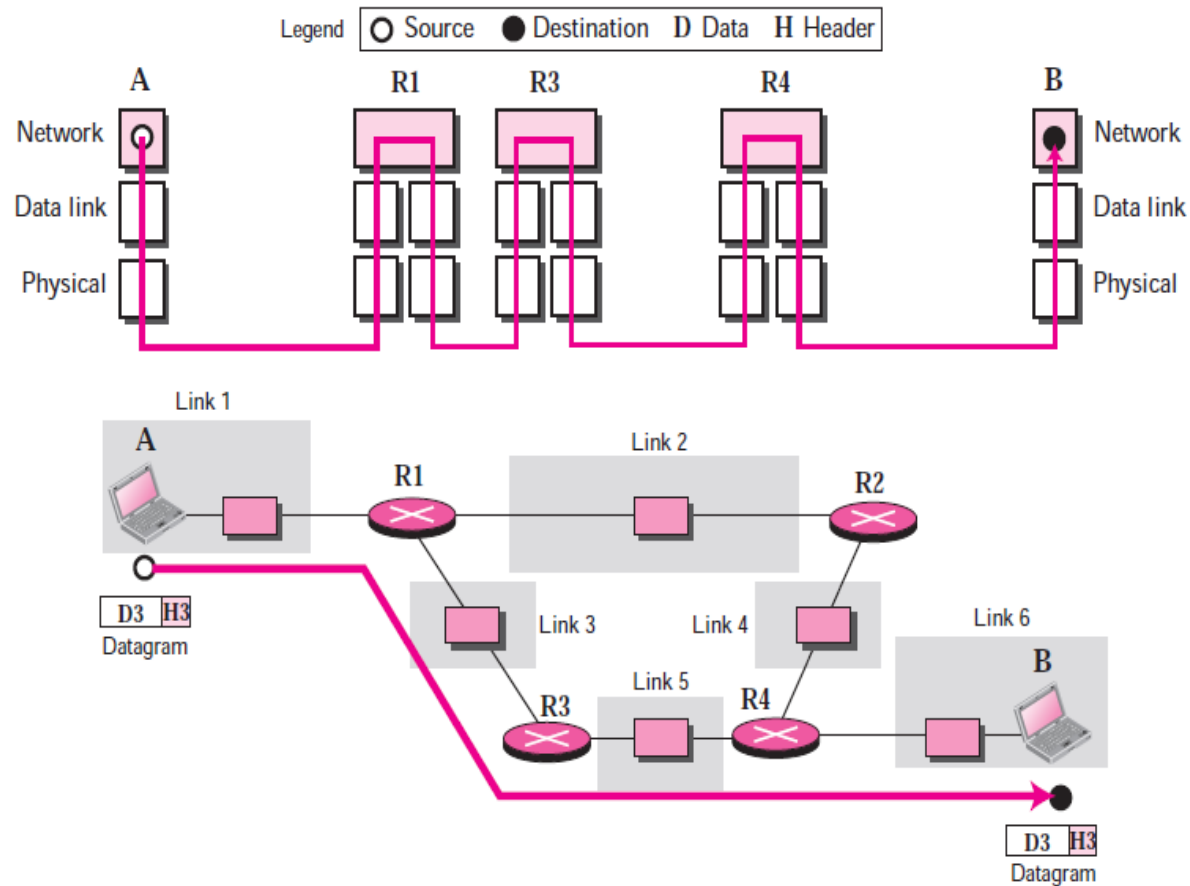
2. Data Link Layer

- TCP/IP does **not define any specific protocol for the data link layer** either
- It supports all of the standard and proprietary protocols
- The communication is also between two hops or nodes
- The unit of communication however, is a packet called a **frame**
- A frame is a packet that encapsulates the data received from the network layer with an added header and sometimes a trailer
- The header, among other communication information, includes the source and destination of frame



3. Network Layer

- At the network layer (or, more accurately, the internetwork layer), TCP/IP supports the Internet Protocol (IP)
- IP is the transmission mechanism used by the TCP/IP protocols
- IP transports data in packets called *datagrams*, each of which is transported separately
- Datagrams can travel along different routes and can arrive out of sequence or be duplicated
- IP does not keep track of the routes and has no facility for reordering datagrams once they arrive at their destination

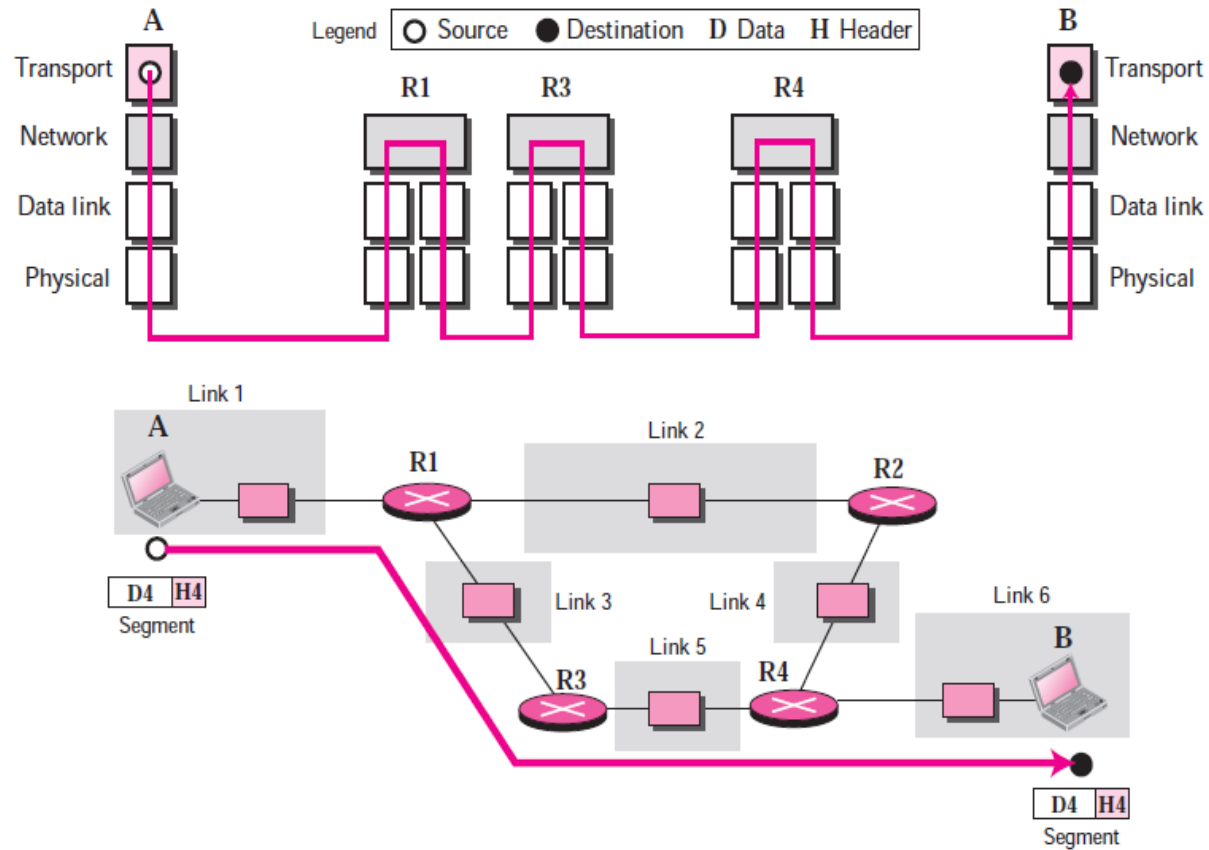


4. Transport Layer

- Only the two end computers need to have the transport layer.
- The transport layer is responsible for delivering the *whole message*, which is called a *segment* from A to B
 - A segment may consist of a few or tens of datagrams.
- As opposed to the network layer which is responsible for sending *individual datagrams*.
- The segments need to be broken into datagrams, and each datagram must be delivered to the network layer for transmission

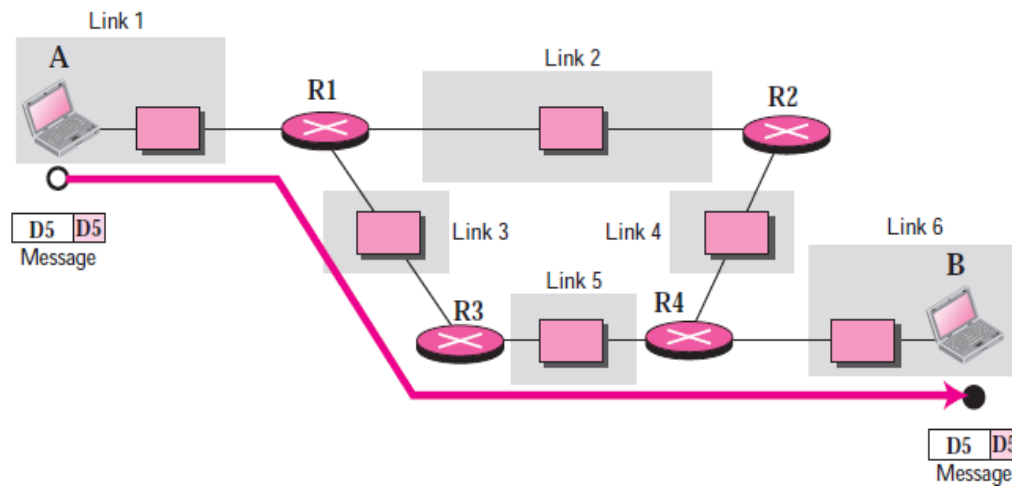
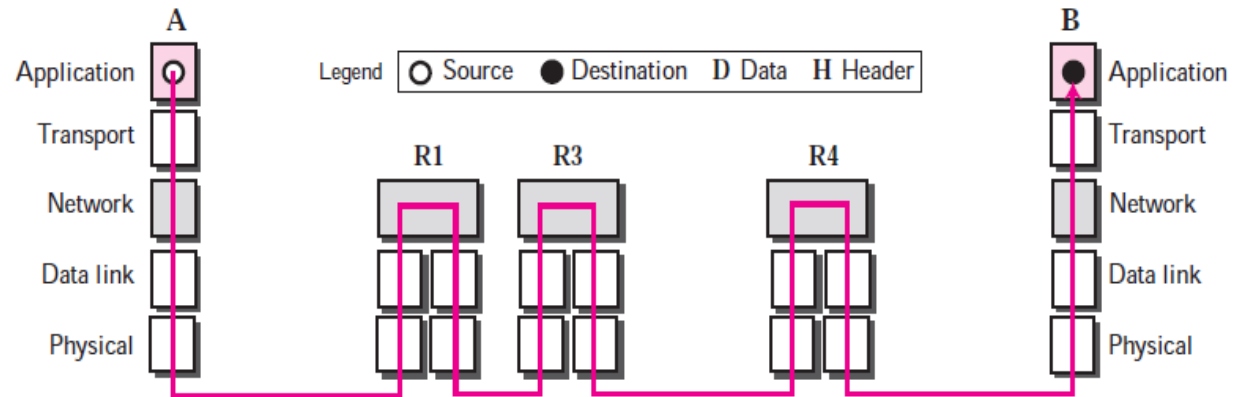


- Since the Internet defines a different route for each datagram, the datagrams may arrive out of order and may be lost
- The transport layer at computer B needs to wait until all of these datagrams to arrive, assemble them and make a segment out of them
- The transport layer was represented in the TCP/IP suite by two protocols:
 - *User Datagram Protocol (UDP)*
 - *Transmission Control Protocol (TCP), (primary)*



5. Application Layer

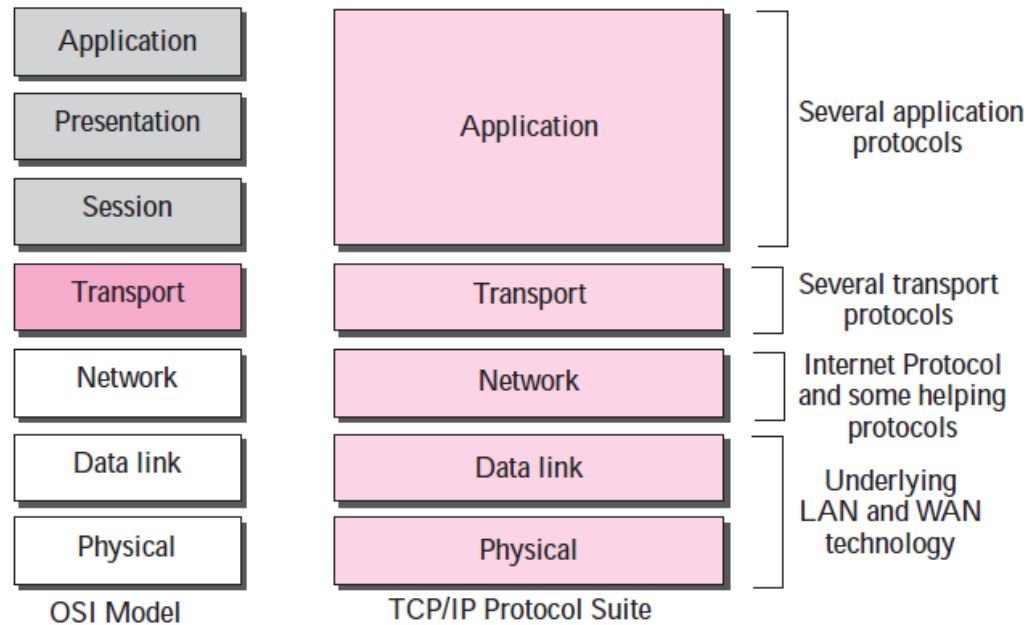
- The application layer in TCP/IP is equivalent to the combined session, presentation, and application layers in the OSI model
- The application layer allows a user to access the services of our private internet or the global Internet
 - electronic mail, file transfer, accessing the World Wide Web, etc.
- The unit of communication at the application layer is a *message*





Difference between OSI & TCP/IP model

OSI Model vs TCP/IP Model





OSI Model vs TCP/IP Model

7	Application Layer	Application Layer	HTTP (Web)	FTP (File Transfer)	SMTP (Email)	DNS (Name Resolution)	DHCP (IP Assignment)	...
6	Presentation Layer							
5	Session Layer							
4	Transport Layer	Transport Layer	TCP (Transmission Control Protocol)		UDP (User Datagram Protocol)			
3	Network Layer	Internet Layer	IP (Internet Protocol)				ICMP (Internet Control Message Protocol)	
2	Data Link Layer	Network Access Layer	ARP (Address Resolution Protocol)	RARP (Reverse Address Resolution Protocol)	Ethernet (IEEE 802.3)	PPP (Point to Point Protocol)	...	
1	Physical Layer		Ethernet (Physical Layer)	Token Ring (Physical Layer)	Coaxial Cable (Physical Layer)	Fiber Optic (Physical Layer)	...	



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That's the only difference ?



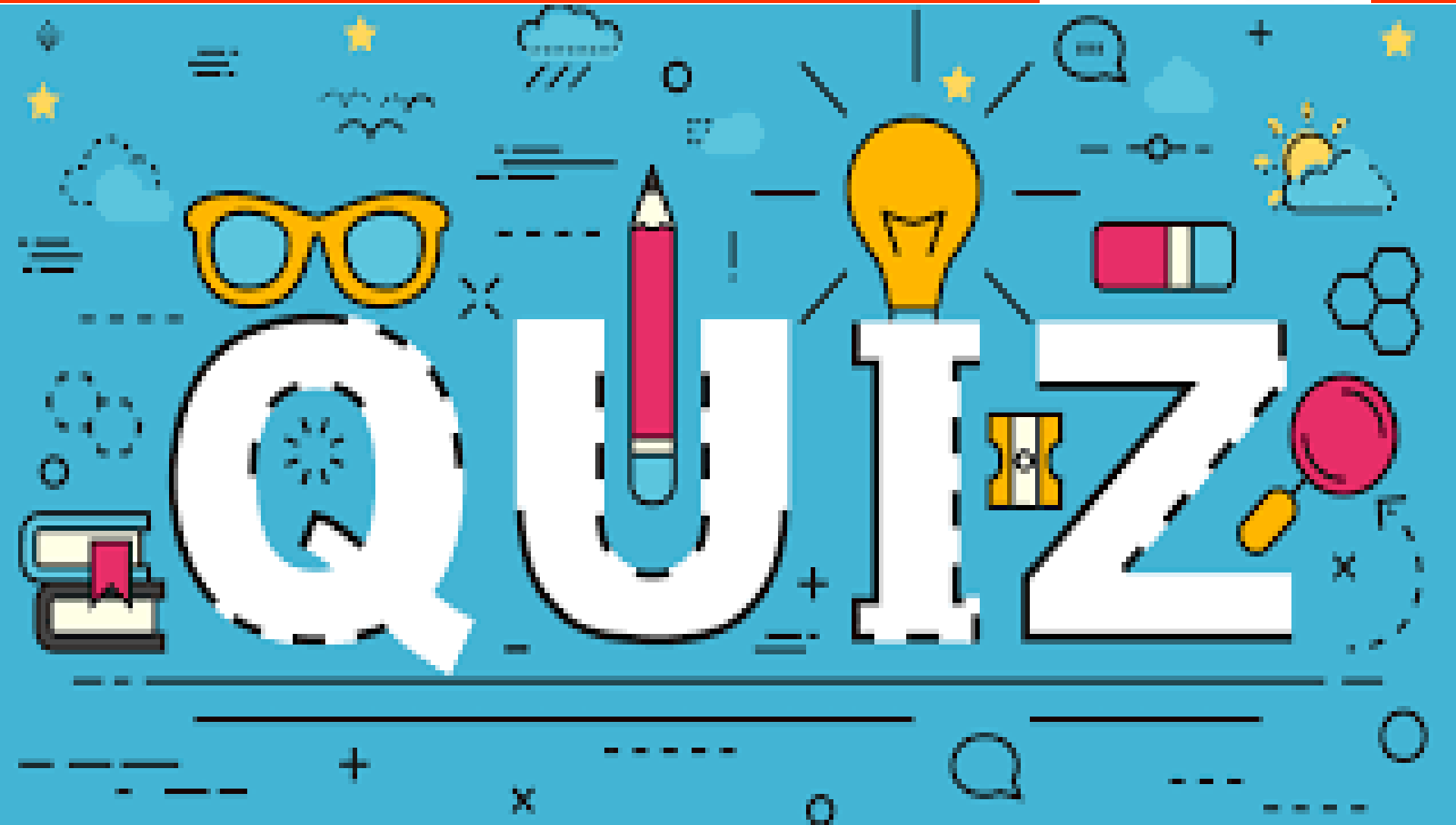
OSI	TCP / IP
Application (Layer7)	Application
Presentation (Layer6)	
Session (Layer 5)	
Transport (Layer 4)	Transport
Network (Layer 3)	Internet
Data Link (Layer 2)	Host to network layer
Physical (Layer 1)	

OSI	TCP / IP
Application (Layer7)	Application
Presentation (Layer6)	
Session (Layer 5)	



Parameter	OSI Model	TCP/IP Model	Why it matters
Nature	A reference model that explains networking concepts.	A protocol suite used to build the Internet.	One is a blueprint, the other is the actual building.
Purpose	Designed to standardize networking concepts.	Designed to solve real communication problems.	Theory vs implementation.
Origin	Developed before practical Internet deployment.	Developed while building ARPANET/Internet.	One was planned first; the other evolved through real use.
Layers	Seven small, specialized layers.	Four broader layers.	TCP/IP merges some OSI layers for simplicity.
Layer Independence	Strict separation of responsibilities.	More flexible; protocols may span responsibilities.	Easier to understand vs easier to implement.
Protocols	Does not prescribe specific protocols.	Defines actual protocols (TCP, IP, UDP, HTTP, DNS, etc.).	OSI says what should happen; TCP/IP says how.
Reliability	Depends on whichever protocols implement the layers.	TCP provides reliable communication; UDP provides fast, unreliable communication.	Reliability is optional depending on the protocol used.
Usage Today	Mainly education, documentation, troubleshooting.	Used by every modern computer network.	Every website, email, and video call uses TCP/IP.
Flexibility	Easier to understand and extend conceptually.	Easier to deploy and interoperate globally.	One is good for learning; the other is good for working.

QUIZ



A customer accesses an online banking website using HTTPS. Before the data is transmitted over the network, it is encrypted to protect sensitive information. Which OSI layer performs this function?

Application Layer

Session Layer

Presentation

Layer

Transport Layer

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A student is attending a Zoom class while simultaneously watching YouTube on the same laptop. Both applications receive data correctly. Which Transport Layer feature makes this possible?

Routing

Port Numbers

MAC Addressing

VLAN Tagging



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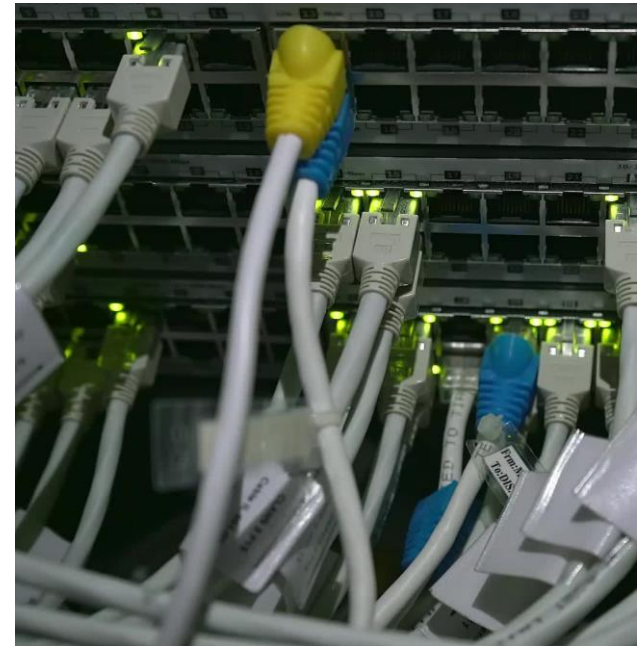
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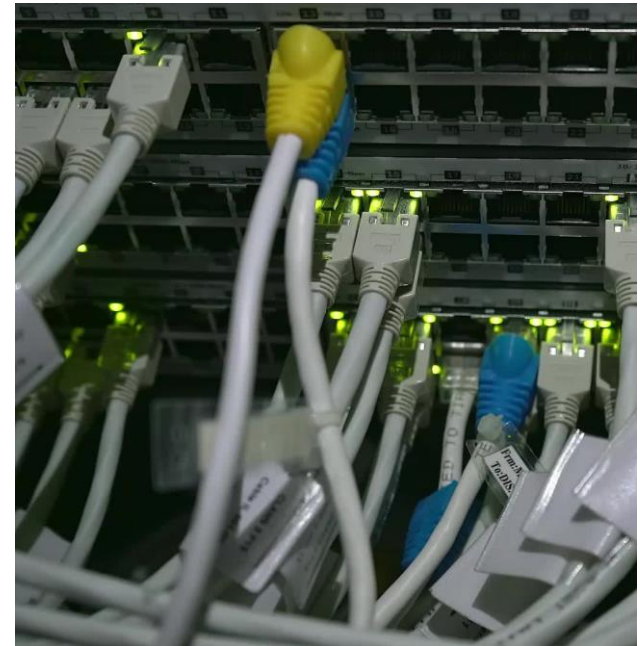
A network administrator notices that a cable has been cut. As a result, no frames, packets, or messages can be transmitted across the network. Which layer is the first to be affected?

- A. Physical Layer
- B. Data Link Layer
- C. Network Layer
- D. Transport Layer



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During a walkie-talkie conversation, only one person can speak at a time, and the communication system manages whose turn it is. Which Session Layer function is involved?

- A. Dialog Control
- B. Compression
- C. Routing
- D. Error Detection



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A student in the Computer Science department of Chitkara University, Rajpura sends data to his friend in Chitkara University, Baddi. Which OSI layer determines the path the data should take?

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- B. Data Link Layer
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A Windows computer sends a document to a Linux server. The Presentation Layer ensures that Unicode characters are interpreted correctly on both systems. Which function is illustrated?



Routing



Encryption



Data Translation



Flow Control

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Routing



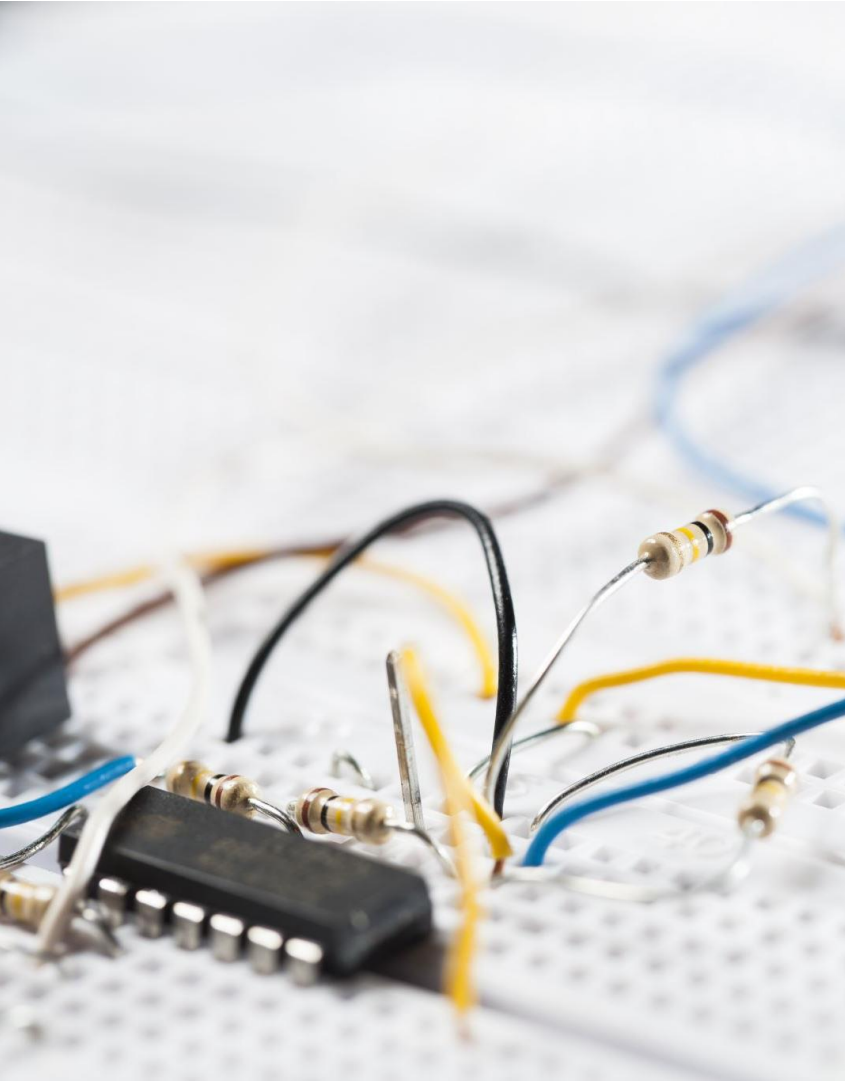
Encryption



Data Translation

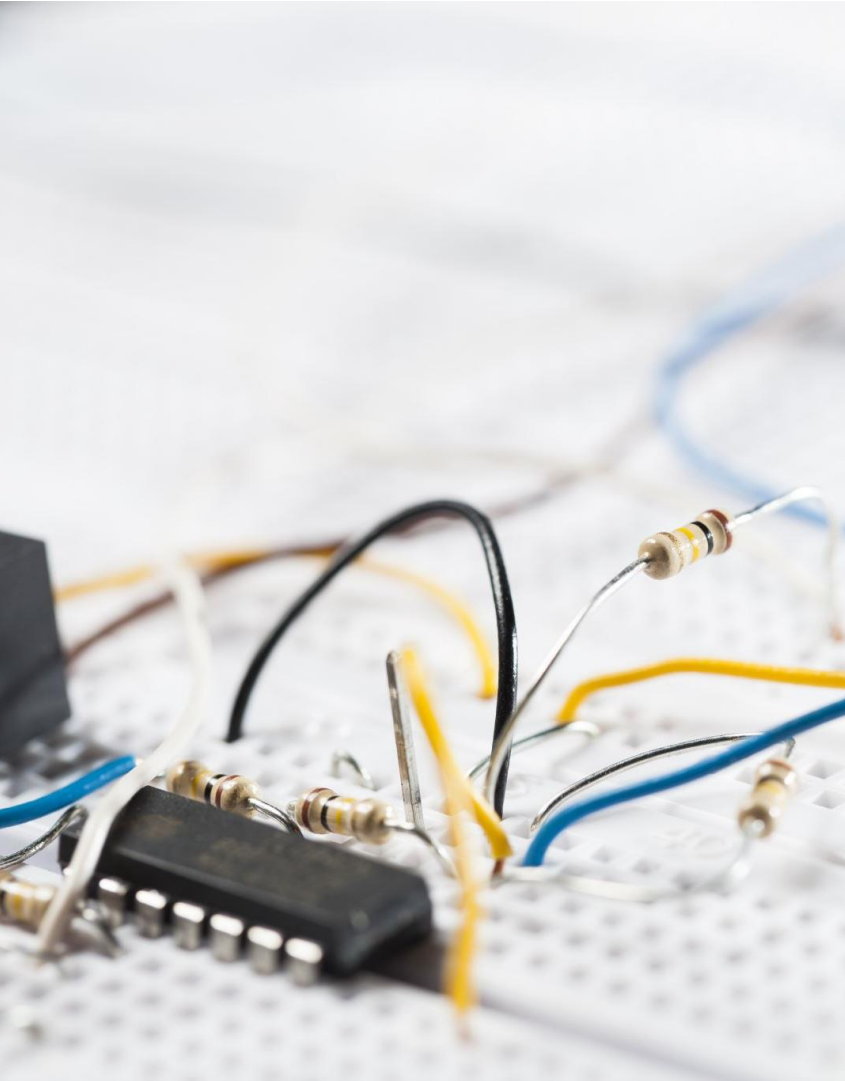


Flow Control



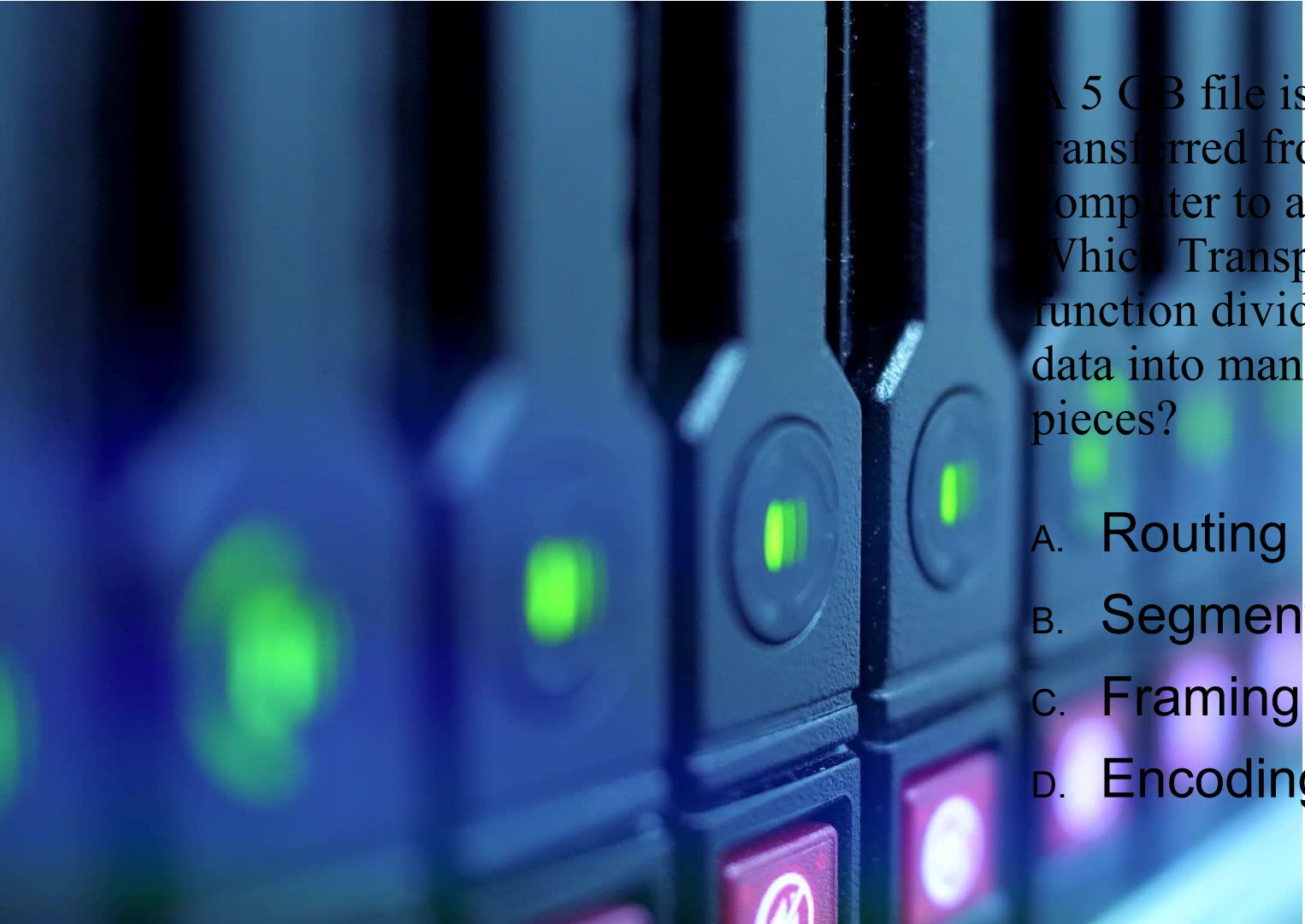
A sender divides a continuous stream of bits into identifiable units before transmission. These units contain source and destination MAC addresses. What are these units called?

- A. Packets
- B. Segments
- C. Frames
- D. Datagrams



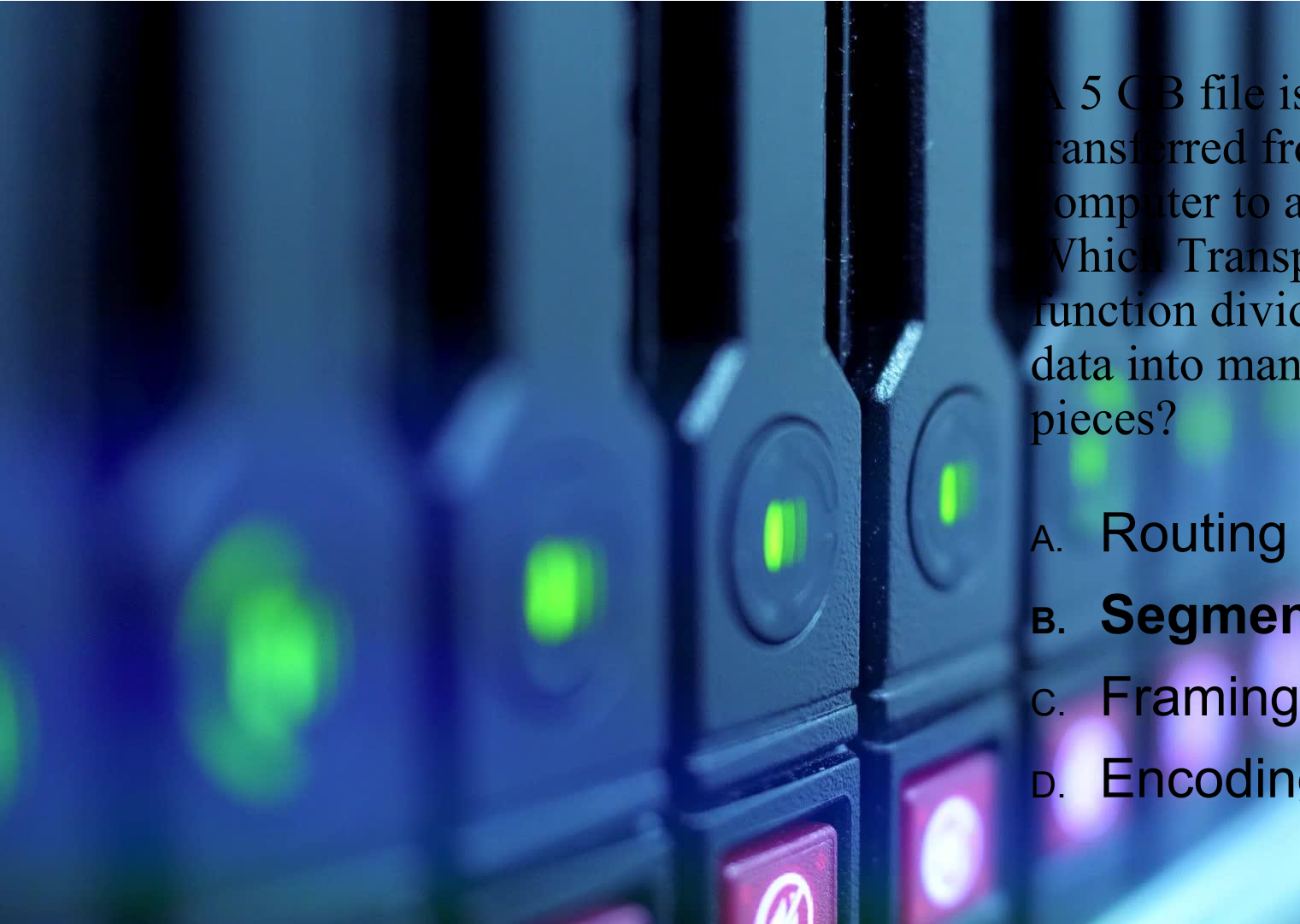
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A 5 GB file is transferred from one computer to another. Which Transport Layer function divides the data into manageable pieces?

- A. Routing
- B. Segmentation
- C. Framing
- D. Encoding

A close-up, slightly blurred photograph of a network switch or patch panel. Several ports are visible, each with a glowing green LED light, indicating active connections. The background is dark and out of focus.

A 5 GB file is transferred from one computer to another. Which Transport Layer function divides the data into manageable pieces?

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- B. **Segmentation**
- C. Framing
- D. Encoding



A browser displays: “404 Not Found”.
Which Application Layer protocol generated this response?

- A. TCP
- B. IP
- C. HTTP
- D. Ethernet



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Which Application Layer protocol generated this response?

- A. TCP
- B. IP
- C. HTTP**
- D. Ethernet



A
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- A. Data Link Layer
- B. Network Layer
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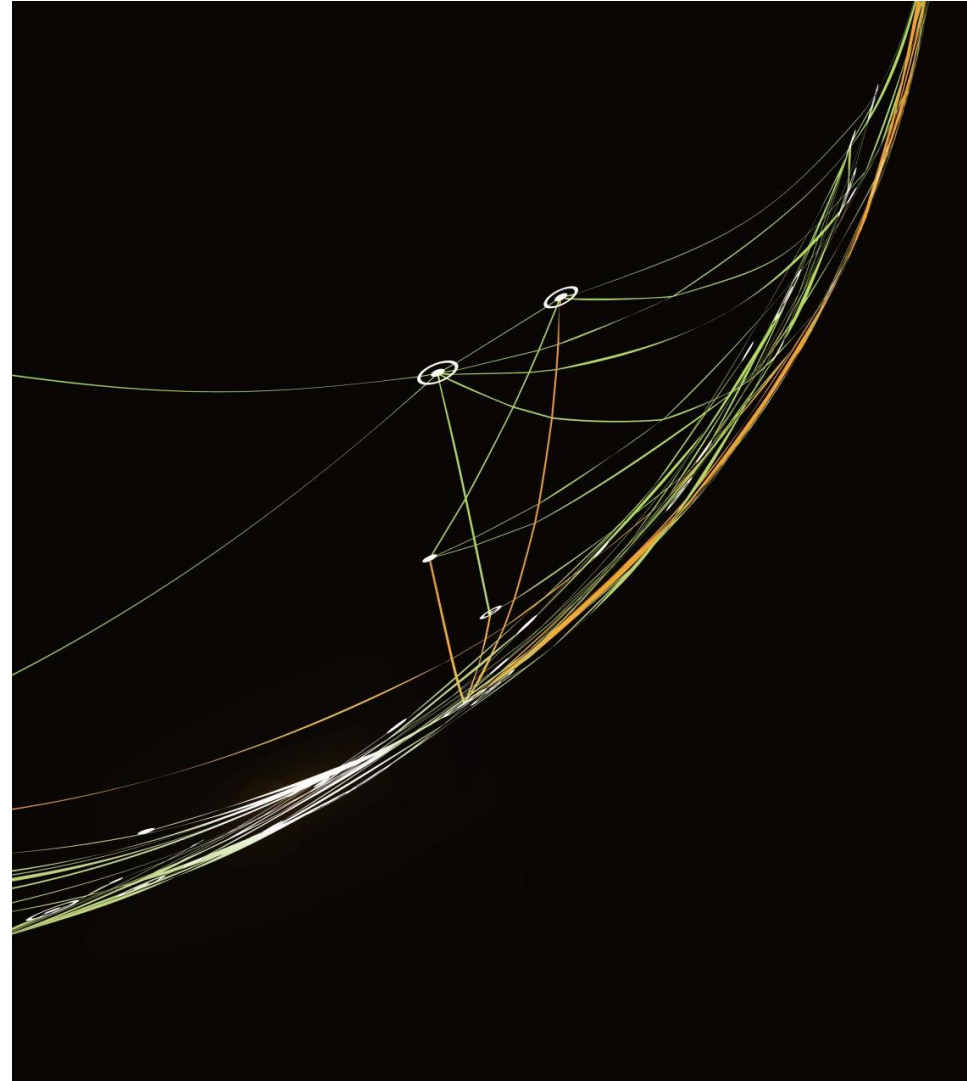


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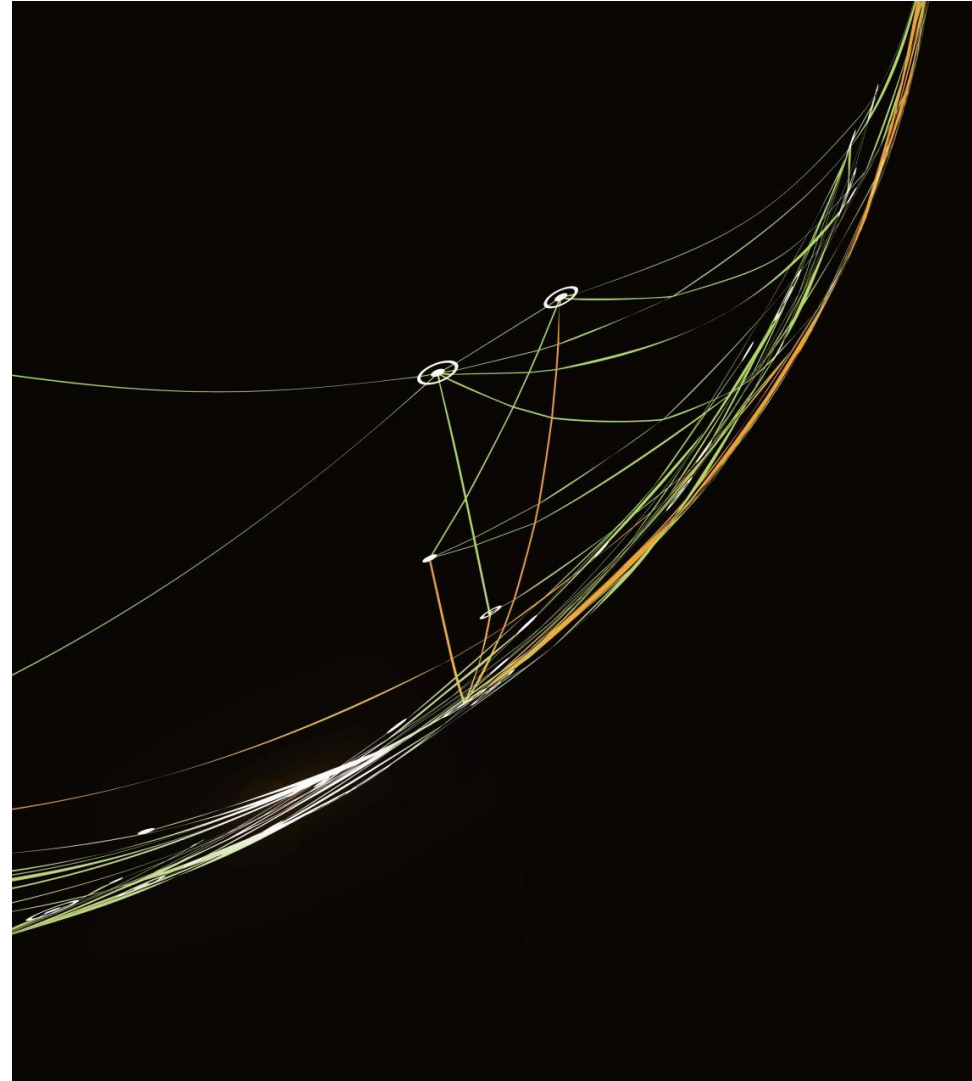
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A 100 GB database backup is being transferred. Every 10 GB, the transfer records a checkpoint. If the network fails after 70 GB, the transfer resumes from the last checkpoint instead of starting again. Which Session Layer function is demonstrated?

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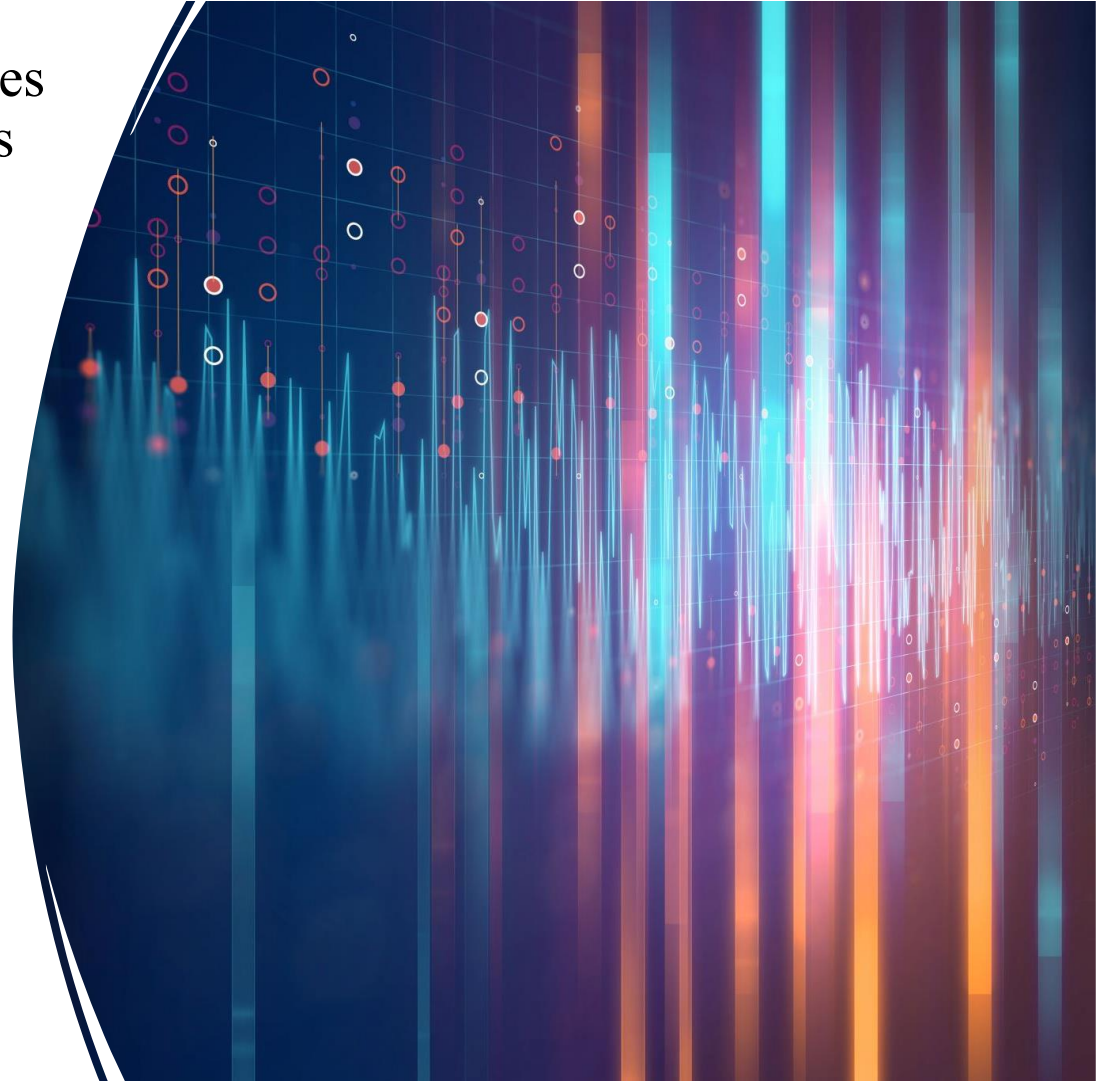
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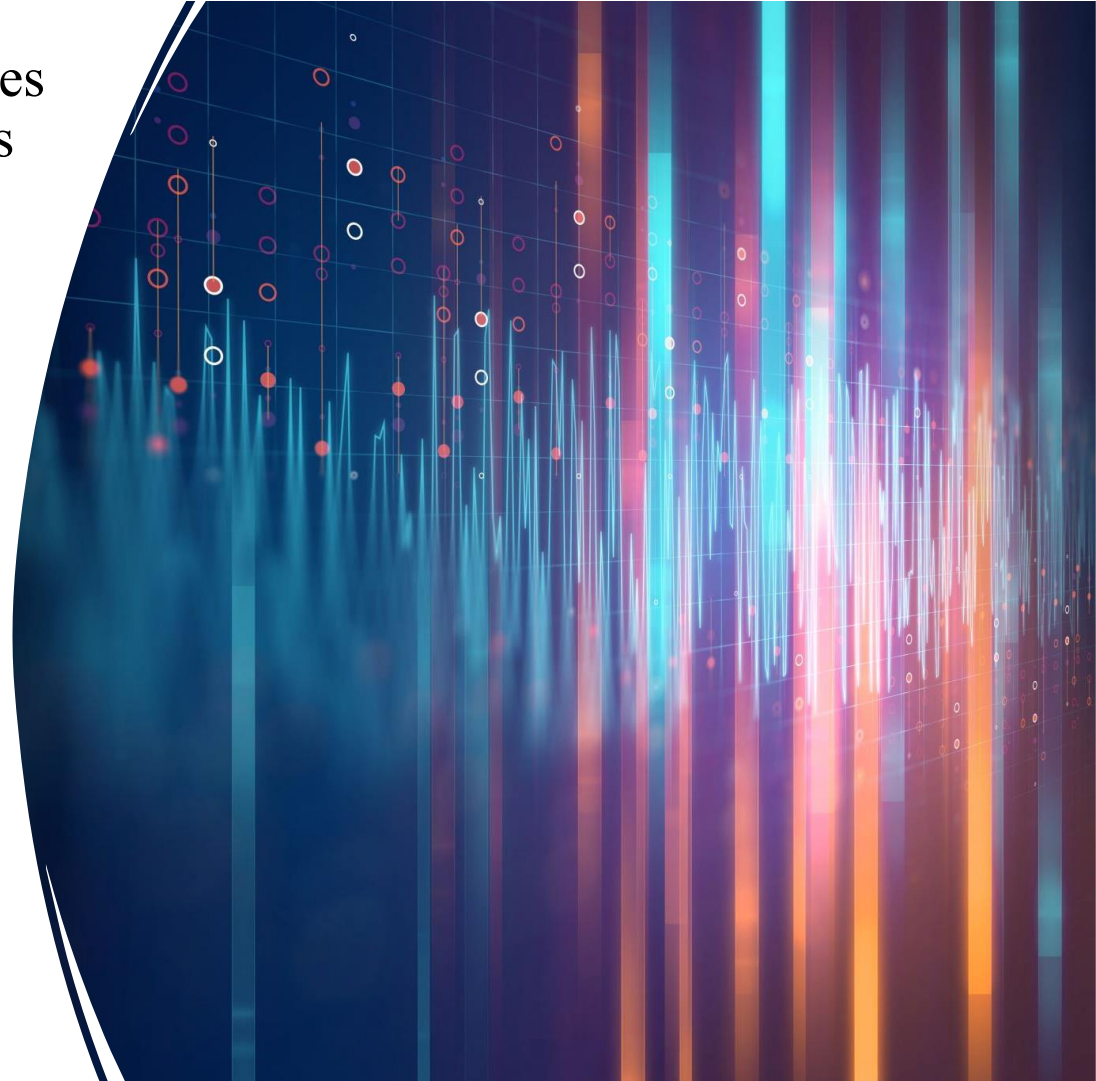
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A frame successfully reaches the destination computer, but the IP packet inside contains an incorrect destination IP address and is discarded.

Which layer successfully completed its job?

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Data Link Layer

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An administrator configures a network interface to operate at 1 Gbps instead of 100 Mbps. Which OSI layer functionality is being modified?



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Physical Layer



Application

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- B. It identifies the destination network.
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Data Link



Network Layer



Transport



Physical Layer



Data Link



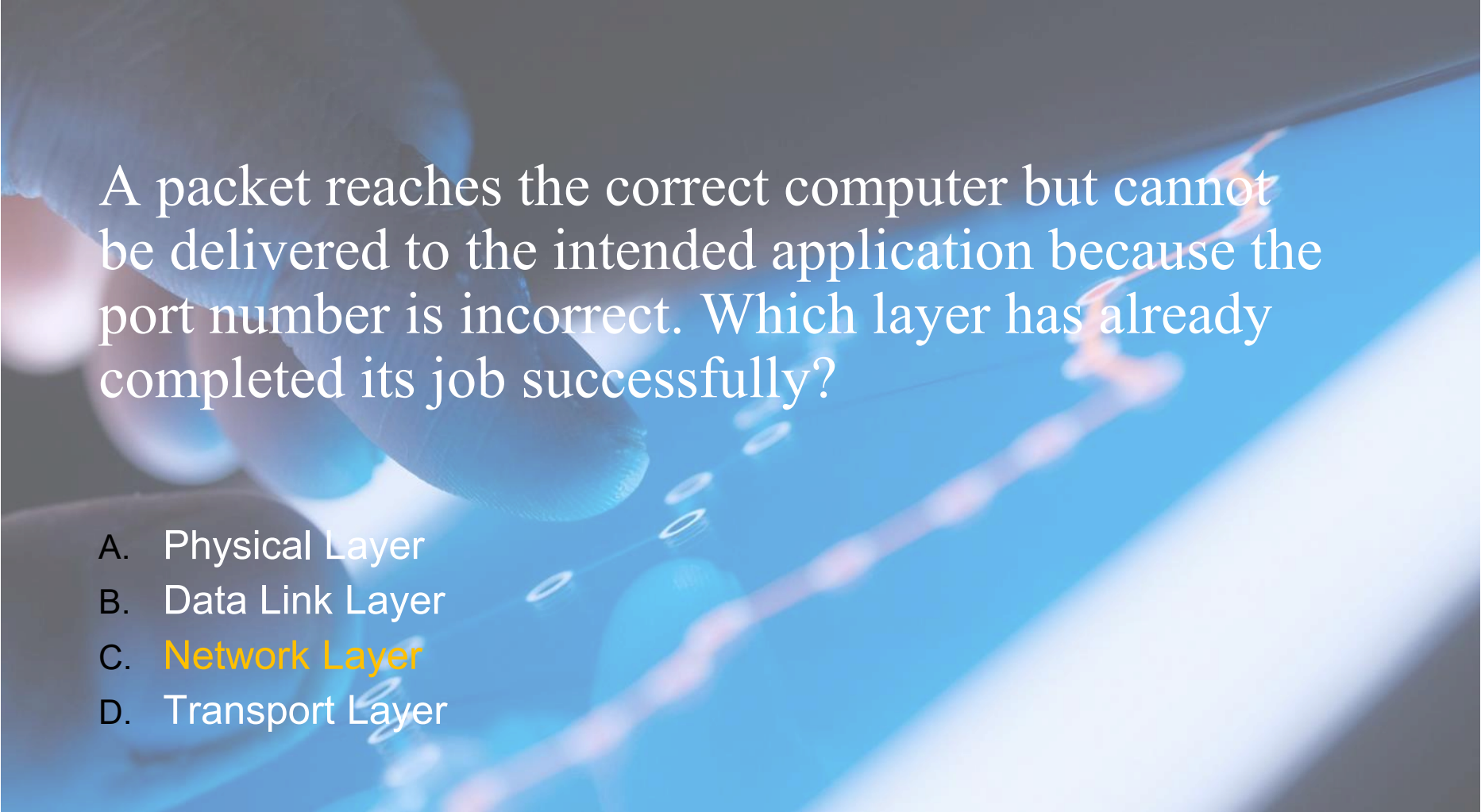
Network Layer



Transport

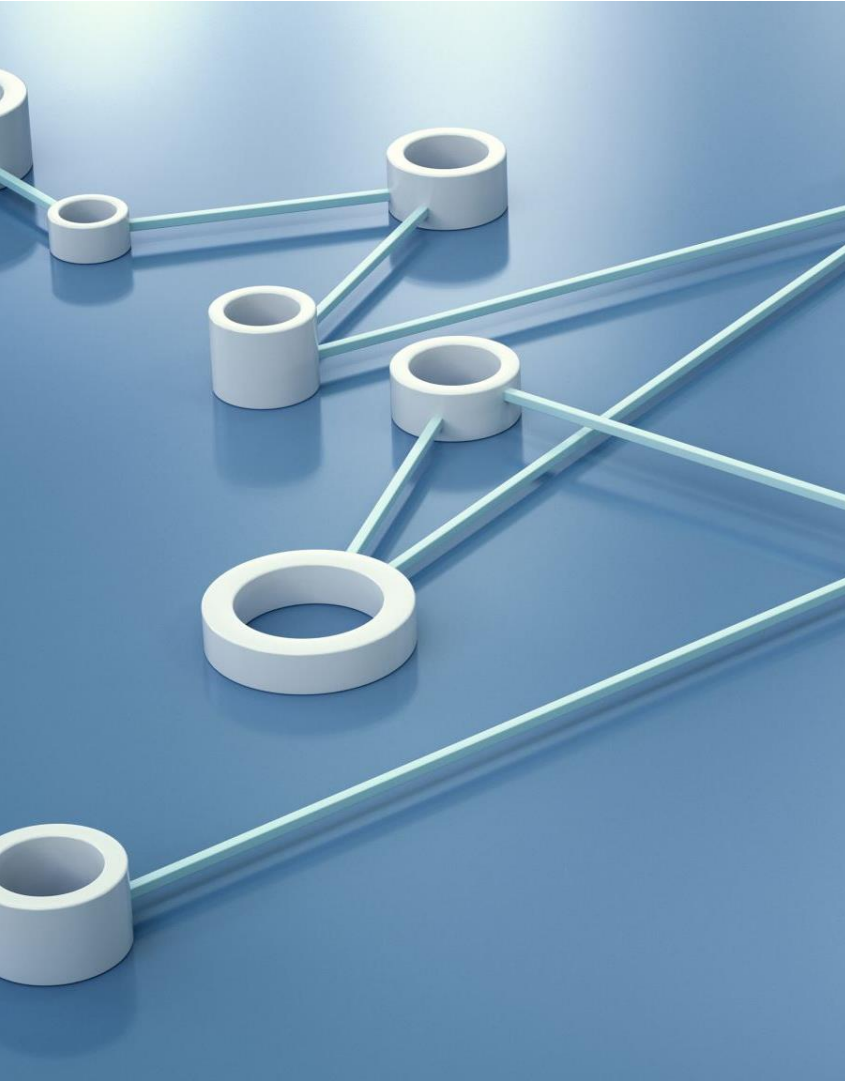
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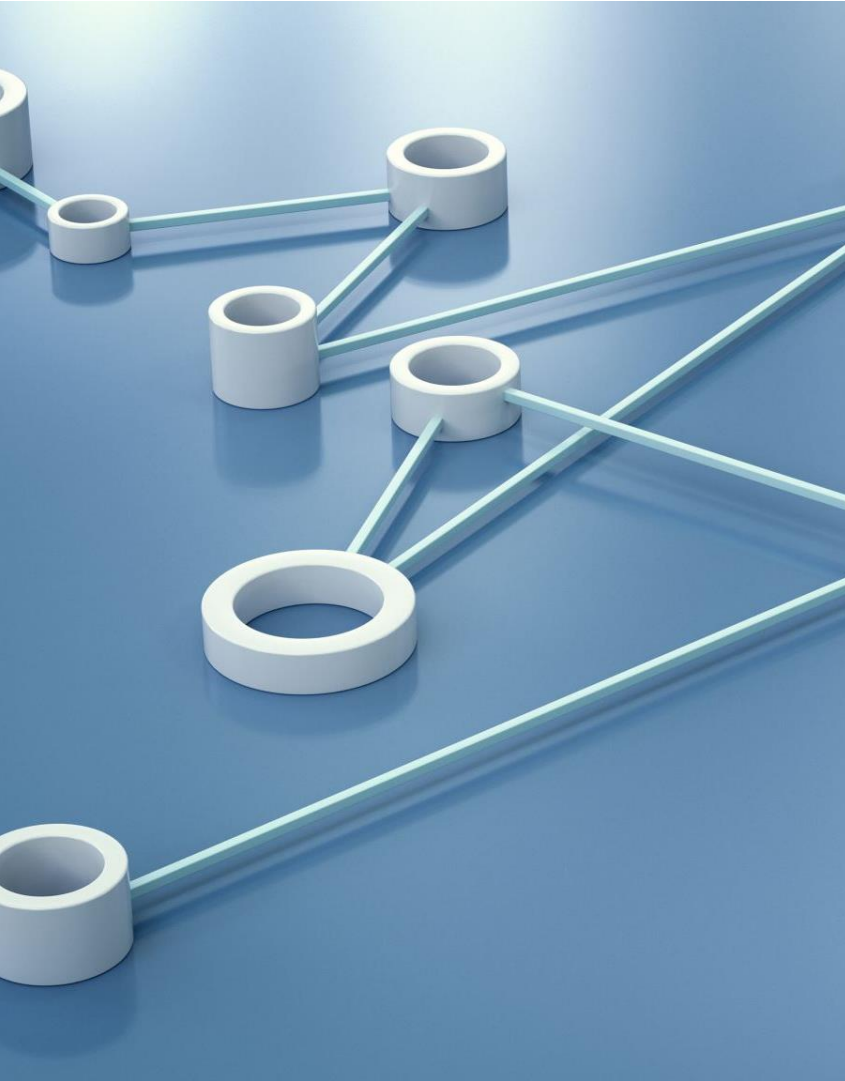
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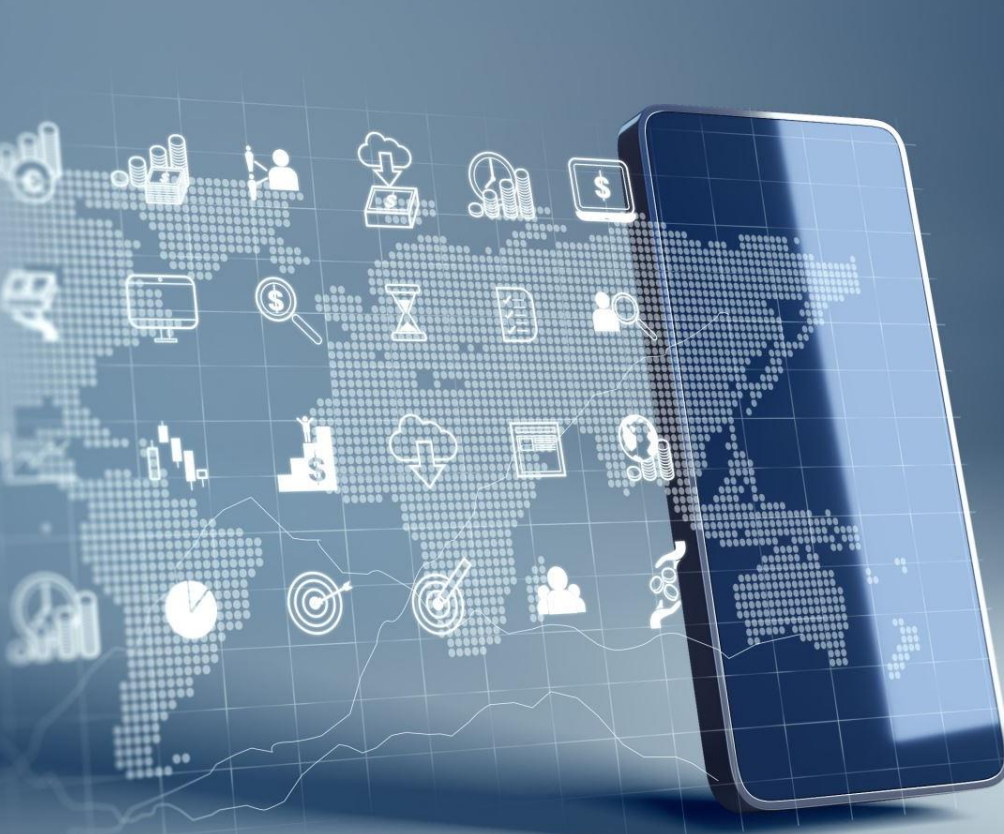
A customer logs into an online banking website. After 10 minutes of inactivity, the website automatically logs the user out to prevent unauthorized access. Which OSI layer is primarily responsible for managing this login session?

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A user accesses a website hosted in another country. The data first travels through the Internet and then reaches the browser. The browser layer solved the problem of locating the correct computer?

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A user types: www.youtube.com; Instead of remembering: 142.250.xxx.xxx
Which Application Layer service performs this conversion?

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- B. FTP (File Transfer Protocol)
- C. DNS (Domain Name System)**
- D. SMTP (Simple Mail Transfer Protocol)



Thank You !